INCH-POUND

MIL-C-44356
23 July 1990

MILITARY SPECIFICATION

COAT AND TROUSERS, CHEMICAL PROTECTIVE, AIRCREW, FLAME RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a coat and trousers that provides chemical protection and flame resistance to military personnel. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The coat and trousers shall be of one type in the following sizes (see 6.2):

Schedule of Sizes for Coat (15 sizes)

	X-Short	Short	Regular	Long
X-Small	X	X	-	-
Small	X	X	X	X
Medium	X	X	X	X
Large	-	X	X	X
X-Large	-	-	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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Schedule of Sizes for Trousers (15 sizes)

	X-Short	Short	Regular	Long
X-Small	-	X	-	-
Small	X	X	X	X
Medium	X	X	X	X
Large	-	X	X	X
X-Large	-	X	X	X

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-50195	-	Thread, Aramid
A-A-50198	-	Thread, Gimp, Cotton, Buttonhole
V-B-871	-	Button, Sewing Hole and Button, Staple (Plastic)
V-F-106	-	Fastener, Slide, Interlocking
JJ-W-155	-	Webbing, Textile (Cotton Elastic)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing
MIL-B-543	-	Buckles, Tongueless and Web Strap
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-T-3530	-	Thread and Twine; Mildew Resistant or Water Repellent Treated
MIL-F-10884	-	Fasteners, Snap
MIL-C-12189	-	Cloth, Coated, Butyl Coated, Toxicological Agents Protective
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic

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MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43303 - Cord, Elastic, Cotton
MIL-T-43566 - Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
MIL-T-43709 - Tape, Textile, Aramid
MIL-C-43858 - Cloth, Laminated, Nylon Tricot Knit, Polyurethane Foam Laminate, Chemical Protective and Flame Resistant
MIL-T-44100 - Thread, Para-Aramid, Spun, Intermediate Modulus
MIL-C-44273 - Cloth, Laminated, Waterproof, Moisture Vapor Permeable and Flame Resistant (4.6 oz.)
MIL-P-44406 - Pouches, Plastic Film
MIL-C-83429 - Cloth, Plain and Basket Weave, Aramid

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials
FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882 - Test Method for Tensile Properties of Thin Plastic Sheeting

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from this document may appear in the sample, in which case this document shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Outer shell. The outer shell material shall conform to MIL-C-44273. As an alternate, used only on designated components of coat and trousers, the outer shell material shall conform to type II, class 1 of MIL-C-83429 (see 3.3.6.2, 3.3.8.2 and 3.5.1).

3.3.2 Lining. The lining material shall conform to type II of MIL-C-43858.

3.3.3 Butyl cloth. The butyl cloth used for protective pieces beneath all coat and trouser pockets shall conform to MIL-C-12189. The heavily coated side of the butyl cloth, when used, shall face to the outside of the coat or trousers. Initial toxicological agents resistance shall meet the requirements specified in MIL-C-12189 when tested in accordance with 4.4.1.1. The colorfastness and dry crocking requirements shall not apply.

3.3.4 Elastic cord. The cotton elastic cord for the drawcord at bottom of coat shall be 3/16 inch in diameter, conforming to class 2 of MIL-C-43303. The cotton elastic cord for the cone closure shall be 3/32 inch in diameter, conforming to class 2 of MIL-C-43303. The color shall be a close approximation to the outer shell fabric and the colorfastness requirements shall not apply.

3.3.4.1 Elastic cord resin impregnating agent. The resin agent used for dipping or impregnating the ends of the drawcords shall be cellulose acetate or cellulose acetate butyrate to prevent ravelling. The dipped or impregnated end shall be at least 1/2 inch in length.

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3.3.5 Cotton elastic webbing. The webbing for the cone opening located inside coat and fastener tabs located at curtain shall conform to type I, classes 6 or 7 of JJ-W-155.

3.3.6 Slide fastener. The slide fasteners listed below shall conform to size MS of V-F-106, except the fabric for the fastener tape shall be woven from 100 percent spun aramid fiber and have a black oxidized finish to the brass chain.

Coat - Type IV, Style 8
Trouser thigh pocket - Type I, Style 3 (length - 9-1/2 inches)
Trouser leg opening - Type I, Style 3 (length - 13 inches)

Size	Length	Trouser Fly (inches)	Coat Front (inches)
X-Small	XS	-	18
X-Small	S	7-1/2	19-1/2
Small	XS	7-1/2	18-1/2
Small	S	8	20
Small	R	8-1/2	21-1/2
Small	L	9	23
Medium	XS	8	19
Medium	S	8-1/2	20-1/2
Medium	R	9	22
Medium	L	9-1/2	23-1/2
Large	S	9	21
Large	R	9-1/2	22-1/2
Large	L	10	24
X-Large	S	9-1/2	-
X-Large	R	10	23
X-Large	L	10-1/2	24-1/2

3.3.6.1 Slide fastener tape. The color of the slide fastener tape shall meet Sage Green 1590 colormatching requirements. The colormatching under incandescent lamplight at 2300 $\pm$ 200K shall not apply.

3.3.6.2 Slide fastener pull thongs. Thongs for the slide fasteners shall be made from the outer shell or alternate outer shell material (see 3.3.1). Instructions for cutting and fabrication shall be specified in Table I.

3.3.7 Hook and loop fastener tape. The hook and loop for the coat and trousers shall conform to type II, class 1, green 1565 of MIL-F-21840. The colorfastness requirements shall not apply.

3.3.8 Tape.

3.3.8.1 Adjustment strap tape. The adjustment strap tape for the hip adjustment on the trousers shall be type I, class 3, 5/8 inch width, conforming to MIL-T-43566, except that the colorfastness requirements shall not apply.

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3.3.8.2 Chest pocket pull tapes. The chest pocket pull tapes shall conform to type I, class 3, 3/8 inch width of MIL-T-43709. The color shall be Sage Green 1590 and the colorfastness and curvature requirements shall not apply. As an alternate, the pocket pull tapes shall be made from outer shell or alternate outer shell material (see 3.3.1). Instructions for cutting and fabrication shall be as specified in Table I.

3.3.8.3 Seam sealing tape. The tape for covering and sealing all designated seams and stitching shall be cut in 1 +1/16, -0 inch wide strips of laminated cloth fabricated in layers as specified below:

a. The white outer layer shall consist of microporous expanded polytetrafluoroethylene (PTFE) film weighing 0.6 + 0.2, - 0.1 ounces per square yard.

b. Adhesive, thermoplastic not less than 5.4 mils thick.

3.3.8.4 Curtain tab. The curtain tab tape shall be 1 inch width conforming to MIL-T-43566. The color shall be Sage Green 1590, except colorfastness requirements shall not apply.

3.3.9 Snap fasteners. Snap fastener assemblies conforming to MIL-F-10884 shall be used to secure front closure of trousers. The fasteners shall conform to style 2 or 2A, finish 2. Style 2 shall consist of a 24 line button, size 1 or 2 socket for the female half; and eyelet size 1 or 2, with stud for the male half. Style 2A shall consist of a button, size 1 or 2 and socket for the female half; and eyelet, size 1 or 2, with stud for the male half. The following exception to MIL-F-10884 shall apply: A double stud or as an alternate, style 2 or 2A stud-eyelet combination shall be used on the right side of trousers at waist opening. The barrel length of button and eyelet shall be selected by the contractor so as to ensure a secure clinch of components to material.

3.3.10 Cord closure. The spring-loaded cord closure shall be size C or D, 1/4 inch diameter, Acetal material conforming to ITW Nexus Barreloc TM (see 6.7), or equal (see 6.8).

3.3.11 Buckles. The buckles for the hip adjustment straps shall be tongue-less, 2 bar, conforming to type II, class 3, style 3, 5/8 inch of MIL-B-543.

3.3.12 Gimp. The cotton gimp for reinforcing buttonholes shall conform to type I or II, Ticket No. 8 of A-A-50198, the color shall meet Sage Green 1590 matching requirements and the colorfastness requirements shall not apply.

3.3.13 Buttons. The buttons shall be type II, class D, style 26, 30-line, dull finish, conforming to V-B-871. The color shall meet Sage Green 1590.

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3.3.14 Thread.

3.3.14.1 Para-aramid, spun, intermediate modulus. The para-aramid thread for seaming and stitching the coat and trousers shall conform to MIL-T-44100. The color of the thread shall meet Sage Green 1590 matching requirements and the colorfastness requirements shall not apply. The thread used to sew the laminate cloth for the lining may be white where not exposed on the outside. The aramid thread shall be water repellent treated with a polytetrafluoroethylene (PTFE) finish capable of conforming to MIL-T-3530 initial vertical rise wicking requirements (see 6.6). Lubrication of the PTFE treated thread, by any means in the end item fabricator's plant prior to or during the sewing, is prohibited.

3.3.14.2 Aramid (alternate). As an alternate to the MIL-T-44100, size T-50, thread conforming to A-A-50195, quarpel treated, 600 denier, Sage Green 1590 thread may be utilized.

3.3.15 Labels. All clothing labels shall be in accordance with DDD-L-20. Each coat and trousers shall have an identification, a size, and an instruction label. These three labels may be combined. The colorfastness requirements shall not apply.

3.3.15.1 Identification label. The identification label shall conform to type VI, class 1 of DDD-L-20, except that the surveillance marking requirements in MIL-C-43858 of the laminated cloth shall appear under the last line of the inscription on the label. The surveillance marking may be hand printed or stamped. Size of characters for the lot number shall be a minimum of 10 points affixed with Black (No. 37038) ink conforming to type I of A-A-208.

3.3.15.2 Size label. The size label shall conform to type VI, class 2 of DDD-L-20.

3.3.15.3 Instruction label. The instruction label shall conform to type VI, class 3 of DDD-L-20 and shall include the following instruction information for each respective size. Instruction information shall be printed with letters not less than a minimum 10 point (approximately 1/8 inch).

1. The coat and trousers are to be worn as a duty uniform in place of the standard aramid flightsuit and chemical protective overgarment when chemical protection is required.

2. The coat and trousers are not launderable. They may only be laundered for training purposes.

3. Protection provided by this coat and trousers may be degraded by gross contamination with fuel, oil or hydraulic fluid.

DO NOT REMOVE THIS LABEL

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3.3.15.4 Combination label. Size, identification, and instruction labels may be combined conforming to type VI, class 14 of DDD-L-20, except that the lot number of the laminated cloth shall appear under the last line of inscriptions on the label and may be hand printed or stamped. Size of characters for the lot number shall be a minimum of 10 points affixed with Black (No. 37038) ink conforming to type I of A-A-208.

3.3.15.5 Fit and size prediction legend. The fit and size prediction legend shall be as specified below. The markings shall be applied to the vapor barrier bag in accordance with 5.4.2.

COAT, CHEMICAL PROTECTIVE, AIRCREW, FLAME RESISTANT

TROUSERS, CHEMICAL PROTECTIVE AIRCREW, FLAME RESISTANT

(1) The aircrew chemical protective coat and trousers are sized similarly to the current flightsuit. The coat and trousers are a duty uniform, not an overgarment. The main difference in sizing is the aircrew, chemical protective coat has a larger chest grade. In addition, the aircrew chemical protective coat and trousers are sized adjectively (small, medium, large) as opposed to numerically (38, 40, 42). Use the size prediction chart as a guide to determine your approximate coat or trouser size. If your chest size or waist size falls between two sizes you should go to the larger size. Examples: If the subject's height is 5'10" and has a chest measurement of 40 inches, they should wear a medium-regular coat. If the subject's waist is 34 inches and has an inseam measurement of 32 inches, they would wear a medium-regular trouser.

NOTE: This is a general guide. The coat and trousers will not fit as snugly as the standard flightsuit. Wear the size that is most comfortable.

(2) Belt loops are provided for a belt, if needed, to assist in keeping up the trousers.

(3) CHEMICAL PROTECTION IS PROVIDED FOR UP TO 30 DAYS FROM THE TIME THIS BAG SEAL HAS BEEN BROKEN OPEN. The coat and trousers can be worn for up to 30 consecutive days. Within the first 15 days of use, 24 hours of protection is offered. Up to 30 days of use only 6 hours of protection is offered. At the end of 30 days use and/or after 6 hours use in a chemical environment the coat and trousers are to be replaced.

(4) When maximum protection is required, wear this coat and trousers with the protective mask and hood; glove set, chemical protective; and green vinyl overshoes.

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Coat prediction chart (inches)

X-Small - X-Short Height: Up to 63 Chest: Up to 33	X-Small - Short Height: From 63 to 67 Chest: Up to 33	Small - X-Short Height: Up to 63 Chest: From 33 to 37
Small - Short Height: From 63 to 67 Chest: From 33 to 37	Small - Regular Height: From 67 to 70 Chest: From 33 to 37	Small - Long Height: From 71 to 74 Chest: From 33 to 37
Medium - X-Short Height: Up to 63 Chest: From 37 to 40	Medium - Short Height: From 63 to 67 Chest: From 37 to 40	Medium - Regular Height: From 67 to 70 Chest: From 37 to 40
Medium - Long Height: From 71 to 74 Chest: From 37 to 40	Large - Short Height: From 63 to 67 Chest: From 41 to 44	Large - Regular Height: From 67 to 70 Chest: From 41 to 44
Large - Long Height: From 71 to 74 Chest: From 41 to 44	X-Large - Regular Height: From 67 to 70 Chest: From 45 to 48	X-Large - Long Height: From 71 to 74 Chest: From 45 to 48

Trouser prediction chart (inches)

X-Small - Short Waist: From 24 to 27 Inseam: From 27-1/2 to 29-1/2	Small - X-Short Waist: From 28 to 31 Inseam: From 24-1/2 to 26-1/2	Small - Short Waist: From 28 to 31 Inseam: From 27-1/2 to 29-1/2
Small - Regular Waist: From 28 to 31 Inseam: From 30-1/2 to 32-1/2	Small - Long Waist: From 28 to 31 Inseam: From 33-1/2 to 35-1/2	Medium - X-Short Waist: From 32 to 35 Inseam: From 24-1/2 to 26-1/2
Medium - Short Waist: From 32 to 35 Inseam: From 27-1/2 to 29-1/2	Medium - Regular Waist: From 32 to 35 Inseam: From 30-1/2 to 32-1/2	Medium - Long Waist: From 32 to 35 Inseam: From 33-1/2 to 35-1/2
Large - Short Waist: From 36 to 39 Inseam: From 27-1/2 to 29-1/2	Large - Regular Waist: From 36 to 39 Inseam: From 30-1/2 to 32-1/2	Large - Long Waist: From 36 to 39 Inseam: From 33-1/2 to 35-1/2
X-Large - Short Waist: From 40 to 43 Inseam: From 27-1/2 to 29-1/2	X-Large - Regular Waist: From 40 to 43 Inseam: From 30-1/2 to 32-1/2	X-Large - Long Waist: From 40 to 43 Inseam: From 33-1/2 to 35-1/2

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3.4 Design.

3.4.1 Coat. The coat shall be a hip length design with stand-up collar and full length sleeves with adjustable tab and hook and loop fastener tapes. Each sleeve shall have an outside one-side bellows type pocket, an inner atropine pocket with an outside flap that covers both atropine and bellows pocket. The left sleeve shall have a pencil pocket located on the front side of the bellows type pocket. The coat shall have two side-opening chest pockets, front closure with slide-fastener and intermittent fastener tapes, and an elastic drawcord in the hem. The coat has an inner curtain attached to the liner at the mid section along with two elastic fastener tabs which will attach to the buttons on the trousers.

3.4.2 Trousers. The trousers shall have a slide fastener front opening with protective flap, two bellows type pockets with side slide fastener openings located at the thigh, two bellow type pockets with flaps located at the calf, hip adjustment buckles and straps, and belt loops. Two buttons are located at the waist front for the lower curtain fastener tabs. Each leg shall have a slide fastener located at the lower inseam for closing the gusset opening with an adjustable tab and fastener tapes.

3.5 Patterns. Standard patterns, which provide 1/2 + 1/8, - 0 inch seam allowance for double-lapped and double-stitched seams of coat and trousers, and 3/8 $\pm$ 1/16 inch seam allowance for single needle seams, except where otherwise specified, will be furnished by the Government. The standard patterns shall not be altered in any way and are to be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns.

3.5.1 List of pattern parts. The components of the coat and trousers shall be cut from the materials as specified in accordance with the pattern parts indicated.

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Materials	Nomenclature	Cut Parts
	Coat	
Outer shell	Back	1
	Front	2
	Sleeve	2
	Curtain 1/	1
	Yoke	1
	Waist adjustment tab 1/	4
	Collar	2
	Chest pocket 1/	2
	Sleeve adjustment tab 1/	4
	Sleeve pocket 1/	2
	Atropine pocket flap 1/	2
	Atropine pocket facing 1/	2
	Pencil pocket 1/	1
	Cone 2/	4
Lining	Back	1
	Front	2
	Sleeve	2
	Cone 2/	4
	Collar insert	1
Tape, textile, cotton	Curtain tab	2
Butyl	Atropine pocket	2
	Chest pocket protective piece	2
	Trousers	
Outer shell	Front	2
	Back	2
	Fly protective flap	1
	Calf pocket 1/	2
	Calf pocket flap 1/	2
	Slide fastener cover (lower leg)	4
	Ankle adjustment tab 1/	4
	Thigh pocket	2
	Thigh pocket welt 1/	2
Lining	Front	2
	Back	2
	Fly protective flap	1
Butyl	Calf pocket protective piece	2
	Thigh pocket protective piece	2

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1/ The alternate outer shell material specified in 3.3.1 may be used for these parts.

2/ Utilize the same pattern piece.

3.6 Construction.

3.6.1 Stitches, seams, and stitching. Stitches, seams, and stitching types specified in table I shall conform to FED-STD-751. When two or more methods of seams or stitch types are given for the same operation, any one may be used. Where stitch type 401 is used, the looper, that is the underthread, shall be on the inside of the coat and trousers. The gage for double-lapped and double-stitched seams shall be 1/4 inch. All other seams shall be 3/8 inch.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table I.

3.6.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch and not more than 1/2 inch. Thread breaks, all stitch types, shall be secured by stitching back of the break not less than 1/2 inch and not more than 3/4 inch. Skipped stitches or thread breaks, 401 stitch type, may be repaired using 301 stitch type. Ends of a continuous line of stitching shall overlap not less than 1/2 inch and not more than 3/4 inch.

3.6.4 Buttonholes and eyelets. The buttonholes shall be either straight cut type or the eyelet and taper bar type worked over gimp with the purling on the outside and the end of the buttonhole stitching and gimp securely tacked. Eyelets, when used, shall have an inside diameter of 3/16 inch to 1/4 inch. The bight of the eyelet stitching shall be not less than 1/16 inch, the purling shall be on the outside and there shall be at least four overlapping stitches.

3.6.5 Mends. The following repair methods are acceptable provided that the contractor performs 100 percent in-process examination of the lining and shell for the repairable defects herein. When it is necessary to repair the outer shell or lining for an open seam in excess of 2 inches, the hem shall be opened and the repair made on the inside in conformance with the applicable operation in table I.

Repair procedures and limitations that apply are as follows:

a. Lining: laminate cloth. The laminate cloth used for the lining may not be mended more than two times in any one garment. The mends shall be made by pinching the material face to face where damaged, and stitching from 1/16 to 1/8 inch from end to end, to form a dart, tapered on each end. The length of the mend shall not exceed 1-1/4 inches.

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b. Outer shell: open seams. Repairs to open seams present in the outer shell of coat or trousers may be made by sewing through the outer shell. The length of the repair stitching shall not exceed 2 inches. Repair stitching shall conform to 3.6.1 through 3.6.3, as applicable. Repairs are limited to two on the coat and two on the trousers. All repairs made to the outer shell must be sealed with seam tape, as specified in 3.6.10.

c. Lining. Repairs to the lining may be hand sewn when plies of material are too thick to be machine sewn or the repairable defect is inaccessible by machine. The thread and stitches per inch shall be as specified in table I for the applicable operation.

d. Lining: open seams. Repairs on open seams in the joining seams of the lining shall be made by pinching the open seam together and stitching both plies 1/16 to 1/8 inch from end to end to form a dart tapered on each end. Outer shell material shall not be caught in the stitching and length of repair shall not exceed 2 inches. Repairs are limited to two on the coat and two on the trousers.

3.6.6 Bartacking. Bartacking shall be 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.6.7 Setting of snap fasteners. When not using an automatic snap fastener setter, a hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.6.8 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table I, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government furnished shaper pattern is forwarded, the component shall conform to that pattern.

3.6.9 Thread ends. All 301 stitch and bartack thread ends shall be trimmed to a length of not more than 1/8 inch. All 515 and 516 stitch thread ends shall be trimmed to not less than 1/4 inch but not more than 1/2 inch in length.

3.6.10 Heat sealing. Seams and stitching on outer shell, as indicated in table I, shall be sealed with heat sealing tape on the innerside of the coat and trousers. The entire width of the seam tape shall be hot air sealed over the seam or stitching. The tape shall be applied by blowing high temperature air on the thermoplastic adhesive face of the sealing tape, which is then driven through pressurized rollers, attaching the tape and sealing the seam and stitching. The machine shall operate on 1200 to 3600 watts, 115 or 220 VAC air

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heater with pneumatically activated driven rollers which press the tape onto the seam or stitching being sealed at a maximum of 20 feet per minute. All seam tapes shall be applied without tension and shall be applied so that a minimum of 3/16 inch overlap is on both sides of the sewn seam. Every effort shall be made to avoid tape end joints where the tape passes over taped seams on the coat and trousers. The ends of each taped seam shall be independently secured on a cross-over sealer such that there are no lifted tape ends or corners. The PTFE laminated outer shell material shall not exhibit any thermal reaction such as puckering, melting, or any sign of degradation. All seam tapes shall overlap a minimum of 3/4 inch at joining points. Sealed seams and stitchings shall show no leakage when preproduction and in-process end item tested as specified in 4.4.5 and 4.4.6.

3.6.10.1 Heat sealing repairs. Repairing the heat sealing tape on the coat and trousers by mending, patching or darning is not allowed and at no time is the removal of heat sealing tape permitted. Seam tape may also be used to repair areas where the original tape does not overlap sewn seams by the minimum, 1/8 inch on both sides. The seam dimensions as above shall apply. No more than two layers of seam sealing tape shall be permitted in any one area.

3.7 Manufacturing operations requirements. The coat and trousers shall be manufactured in accordance with the operations listed in Table I. Although the contractor is not required to follow the exact sequence of operations, the contractor is required to stitch the identification labels with surveillance marking already handprinted or stamped (see 3.3.15.1) to the lining (as specified in operations 15 and 42 of table I) immediately upon cutting to ensure correct identification and traceability of the cloth. Any basting or holding stitching may be used providing threads or needle holes are not visible on the finished garments. Each coat and pair of trousers shall be cut and assembled from the same lot of impregnated cloth.

3.7.1 Figures. Figures 1, 2, and 3 are furnished for information only. To the extent of any inconsistencies between the figures and the written document, the written document shall control.

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting a. Cut the coat and trousers in strict accordance with patterns furnished, which show size, directional lines and marks for proper assembly. Unless otherwise specified, the directional lines indicate the warp direction. b. Cut the outer shell and curtain parts from one lot of material except fly protective flap, belt loops, chest pocket pulls, sleeve adjustment tabs, ankle adjustment tabs, coat waist adjustment tabs, curtain reinforcement tape and slide fastener thongs which may be cut from ends. Parts cut from ends shall approximate the shade of the outer shell. c. Cut all parts of the lining (laminated cloth) from one lot of material. The knit side of laminated cloth is the face of the material (see 6.5.2). d. Directional lines on the patterns for fronts and back of the trousers may vary from the warp direction by not more than 2-1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the directional line to the selvage edge of the fabric and the difference between the two measurements shall not exceed 2-1/2 inches. e. Cut the elastic cord for drawcord at bottom of coat in the following lengths (± 1 inch tolerance): Size — Inches (relaxed) X-Small - X-Short (XS), Short (S) — 42						

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd)						

Size	Inches (relaxed)
Small - XS, S, R, Long (L)	48
Medium - XS, S, R, L	52
Large - S, R, L	56
X-Large - R, L	62

f. Cut the belt loops, chest pocket pulls, (see operation 1.i), curtain reinforcement and slide fastener thongs to conform to the measurements specified in operations 4 and 5.

g. Cut the hook and loop portions of the nylon fastener tape in the following lengths:

Coat:

Chest pocket (2 each of hook and loop) 6-1/2 ± 1/8 inches (1 inch wide tape)
Sleeve pocket (4 each of hook) 1-1/2 ± 1/8 inches (1 inch wide tape)
Atropine pocket flap (4 each of loop) 1-1/2 ± 1/8 inches (1 inch wide tape)
Name tag (1 each of hook and loop) 4 ± 1/4 inches (2 inch wide tape)
Collar closure (1 each of hook and loop) 2 ± 1/8 inches (1 inch wide tape)
Coat flap closure (3 each of hook and loop) 2 ± 1/8 inches (1 inch wide tape)
Sleeve adjustment tab (2 each of hook and loop) 4-1/2 ± 1/8 inches (1 inch wide tape)
Waist adjustment tab (2 each of hook and loop) hook portion 5 ± 1/8 inches and loop portion 4-1/2 ± 1/8 inches (1 inch wide tape)

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) Trousers Calf pockets (4 each of hook and loop) 2 ± 1/8 inches (1 inch wide tape) Ankle adjustment tab (2 each of hook) 6 ± 1/8 inches (1 inch wide tape) Ankle adjustment tab (2 each of loop) 5-1/2 ± 1/8 inches (1 inch wide tape) h. Cut the 5/8 inch wide tape for the back hip adjustment straps on the trousers 6-3/4 ± 1/4 inches for all sizes. Cut the 5/8 inch tape for the front hip adjustment strap 3 ± 1/4 inches for all sizes. i. Cut the aramid tape 5 ± 1/4 inches for chest pocket pulls or see operation 1.f as an alternate. j. Cut two each of 1 inch wide elastic webbing 2 ± 1/4 inches for the fastener tab located inside the coat. k. Cut four each of 1 inch wide elastic webbing 3 ± 1/8 inches for the cone opening. l. Cut parts of butyl coated fabric from one lot of material. The parts shall be cut in such a manner as to ensure that the heavily coated side will face the outside of the garment. m. Cut the elastic cord for the cone closure 12 ± 1/2 inches.						

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Replacement of damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages are excluded and replaced with non-defective and properly matched material.						
3.	Marking. a. The outer shell parts of the coat and trousers shall be marked, ticketed, or bundled to ensure a uniform size and shade throughout the garments. Any method of marking the outer shell parts may be used except: 1. Metal fastening devices. 2. Sew-on type marking. 3. Adhesive type tickets which discolor the material or leave traces of paper or adhesive upon removal of tickets. b. The lining material parts of the coat and and trousers shall be marked and bundled to ensure a uniform size for each coat or trousers. Each bundle shall have the surveillance number clearly identified. Only one surveillance number can be used for each coat and trouser. c. The butyl cut parts shall be marked and bundled to ensure, when attaching to the areas to be protected, the heavily coated side shall face toward the outside of the coat or trousers.						

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make seven belt loops, two chest pocket pulls, six slide fastener thongs and curtain reinforcement tape (see operation 1.f).						
	a. Fold stripping in half lengthwise with the raw edges turned in and stitch 1/16 to 1/8 inch from the turned under edge.	301	EFp-2	10-14	T-50	T-35	
	or						
	b. Fold under side raw edges with edges abutted or overlapped and double-stitch the length of the stripping, with a 1/4 inch gage, using a cover stitch on the underside.	406	EFh-1	10-14	T-50	T-35	
	c. The loops and thongs shall finish 3/8 to 1/2 inch in width.						
	d. The chest pocket pulls shall finish 3/8 to 1/2 inch in width and 5 ± 1/4 inches in length.						
	e. The curtain reinforcement tape shall finish 3/8 to 1/2 inch in width and long enough to cover the entire width of the curtain.						
5.	Attach thongs to slide fastener pulls on coat front, trouser fly, thigh pockets and gusseted trouser closures (see operation 4).						
	a. Cut the thongs 4 ± 1/4 inches in length.						
	b. Thread one end of thong through end of slide fastener pull and with ends even, secure the ends with a bartack.	Bartack		28 per bartack	T-50	T-35	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Attach thongs to slide fastener pulls on coat front, trouser fly, thigh pockets and gusseted trouser closures (see operation 4). (cont'd)						
	NOTE: All stitching through outer shell only, including seams, pockets and adjustment tabs must be sealed with seam sealing tape on the inside of the outer shell with tape specified in 3.3.8.3 and 3.6.10.						
	Make Coat						
6.	Make collar.						
	a. Stitch hook fastener tape to undercollar, as indicated on pattern, 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
	b. Stitch loop fastener tape to top collar, as indicated on pattern, 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
	c. Position inner collar lining piece to under-collar so that the foam side of the lining faces the PTFE laminate side of the undercollar. With raw edges even, stitch along top and side edges.	301	SSe-2(a)	8-10	T-50	T-35	
	d. Position top collar and the undercollar assembly, face to face, and stitch along top and side edges. Turn, work out edges and top stitch 1/4 ± 1/16 inch from top and side edges.	301	SSe-2(b)	8-10	T-50	T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make chest pockets.						
	a. Turn front edges of pocket to inside and place loop portion of nylon fastener tape on inside of pocket. Fold a piece of 1/2 inch aramid tape or alternate tape specified in 3.3.8.2 in half lengthwise (for chest pocket pull) and place between fastener tape and pocket, at the midpoint of the fastener tape lengthwise. Stitch completely around the fastener tape, 1/8 to 3/16 inch from all edges. NOTE: When the operation is completed, the pocket pull shall be caught securely in the two long rows of stitching attaching the fastener tape to the pocket.	301	LSbj-1	10-14	T-50	T-35	
	b. Place a bartack over the pocket pull between the two long rows of stitching attaching the fastener tape.	Bartack		28 per bartack	T-50	T-35	
8.	Attach chest pocket protective piece to outer shell.						
	Position chest pocket protective piece with heavily butyl coated surface facing outward on the finished garment, on front outer shell where indicated on pattern, and stitch 1/16 to 1/8 inch from all edges. Place hook portion of nylon fastener tape over protective piece, positioned as indicated on front pattern and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Attach chest pockets to front outer shell.						
	Position chest pockets over the chest pocket protective piece matching hook and loop portion of nylon fastener tape. Turn under 3/8 inch and stitch the remaining front pocket side, as well as all other pocket sides 1/16 to 1/8 inch from turned under edges. Place a horizontal bartack at each end of the pocket opening. The protective piece shall not extend beyond the sides and bottom edges.	301 Bartack	LSd-1	10-14 28 per bartack	T-50 T-50	T-35 T-35	
10.	Attach name tag.						
	a. Position 2 inch wide loop fastener tape above left chest pocket, as indicated on pattern. Stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
	b. Attach hook fastener tape to the loop portion.						
11.	Attach fastener tape to coat for waist adjustment tabs and front closure.						
	Position loop fastener tapes on right front coat outer shell as indicated on pattern. Position loop fastener tape to coat back outer shell (for waist adjustment tabs), as indicated on patterns. Stitch tapes 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
12.	Make and attach waist adjustment tabs.						
	a. Stitch waist adjustment tabs, face to face 1/8 to 3/16 inch from all edges, keeping short straight end open. Turn, work out edges and stitch 1/4 ± 1/16 inch from all edges.	301	SSe-2	10-14	T-50	T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Make and attach waist adjustment tabs. (cont'd)						
	b. Position hook fastener tape on waist adjustment tab, as indicated on pattern. Stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
	c. Position raw edge of the waist adjustment tabs on coat front outer shell as indicated on the pattern and stitch 1/8 to 3/16 inch from raw edges. Turn the tab back and raise stitch 3/16 to 1/4 inch from the turned edge, burying the raw edges within the seam.	301	LSbk-2	10-14	T-50	T-35	
13.	Construct bi-swing on coat back.						
	a. Fold under pleat on bi-swing, with shell fabric face to face, as indicated on pattern. Stitch 1/16 to 1/8 inch from edge.	301	OSf-1	10-14	T-50	T-35	
	b. Fold upper pleat on bi-swing with shell fabric wrong side to wrong side, as indicated on pattern. Stitch 1/16 to 1/8 inch from edge.	301	OSf-1	10-14	T-50	T-35	
14.	Attach yoke to coat front and back.						
	a. Join yoke to coat front with double-lapped, double-stitched seam with yoke overlapping coat front.	301	LSc-2	10-14	T-50	T-35	
	b. Join yoke to coat back with double-lapped, double-stitched seam with yoke overlapping coat back.	301	LSc-2	10-14	T-50	T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Labels.						
	Sew identification, size, and instruction label, or combination label 1/8 to 3/16 inch from the edge on all four sides to the knit side of lining on right front with bottom edges 1-1/4 ± 1/4 inches from bottom edges of front and positioned as follows:	301	LSbj-1 and SSa-1	8-10	T-50	T-35	
	(1) Size or combination label - Centered (± 1/2 inch) on right front.						
	(2) Identification label (when used) - Positioned to the left of the size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching.						
	(3) Instruction label - Positioned to the right of the size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching.						
	NOTE: Final row of stitches on overlapped edges may be continued to bottom of lining.						
16.	Make neck pleat.						
	Fold back lining in half, knit side to knit side and stitch down 2 inches from neck with a 1/2 inch seam, as indicated on pattern. Fold pleat to one side and stitch across pleat at neck edge.	301 or 401	SSa-1	8-10	T-50 T-50	T-35 T-35	
17.	Join shoulder seams of lining.						
	Position lining fronts and backs face to face and join shoulder seams.	301 or 401	SSa-1	8-10	T-50 T-50	T-35 T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Attach fastener tapes to lining.						
	Position hook fastener tapes to face of lining of left front (along center, front closure). Stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
19.	Attach collar.						
	a. Position lining to outer shell, face to face, with fronts and neck edges even. Insert neck edge of finished collar, with notches even between outer shell and liner, and join plies along front edges, with the stitching continued along entire neck edge, including collar.	301 or 401	SSe-2(a)	8-10	T-50 T-50	T-35 T-35	
	b. Turn coat right side out, work out edges, and edge stitch fronts 1/4 to 5/16 inch from front edges with the stitching continuing across the entire neck edge 1/4 to 5/16 inch from collar seam.	301	SSe-2(b)	8-10	T-50	T-35	
20.	Make atropine pocket flap.						
	a. Position loop portion of nylon fastener tape on the atropine pocket flap as indicated on pattern and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
	b. Fold the atropine pocket flap lengthwise and stitch a 1/4 inch seam on each end. Turn the flap working out the corners and place a row of stitching 1/4 ± 1/16 inch from edges around the flap.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b) and OSf-1	10-14 10-14 10-14	T-50 T-50 T-50	T-35 T-35 T-35	
	c. Overedge the raw edges of the pocket flaps.	301	SSa-1 Class 500	10-14	T-50	T-35	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Make sleeve pocket.						
	a. Turn top edge of pocket to inside, as indicated on pattern, and with the raw edge turned in, single stitch 1/16 to 1/8 inch from edge. The hem shall finish 1 ± 1/8 inches wide.	301 or 401	EFb-1	10-14	T-50 T-50	T-35 T-35	
	b. Position hook portion of the nylon fastener tape, as indicated on pattern, and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	T-50	T-35	
	c. Turn top edges of pencil pocket to inside, 3/8 ± 1/16 inch and with raw edges turned in, stitch 3/16 to 1/4 inch from turned in edge. NOTE: The pencil pocket shall be on the left sleeve pocket.	301	EFb-1	10-14	T-50	T-35	
	d. With side edges turned in 3/8 ± 1/16 inch, stitch pencil pocket to left sleeve pocket folding in bottom pleats as indicated on pattern. Stitch 1/16 to 1/8 inch from turned in edges. Stitch down middle of pencil pocket carrier (forming two separate pencil slots).	301	LSd-1	10-14	T-50	T-35	
	e. As indicated on the pattern, fold to form the pocket bellows corner and stitch with a 1/4 inch seam.	301	LSb-1	10-14	T-50	T-35	
	f. As indicated on pattern, fold both the side and bottom with the wrong sides together and raise stitch 1/8 to 3/16 inch from edges. Fold face to face to form bellows on the side and bottom and raise stitch 1/8 to 3/16 inch along the side and bottom.	301	OSf-1	10-14	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Attach sleeve pocket to the outer sleeve.						
	Position sleeve pocket on the sleeve with bellows facing rearward, and turning in the raw edges, stitch along the edges with a 1/8 ± 1/16 inch raised stitch. Place a bartack at top corner (bellows side).	301 Bartack	I.Sd-1	10-14 28 per bartack	T-50 T-50	T-35 T-35	
23.	Make atropine pocket and opening.						
	a. Position butyl pocketing on sleeve liner and stitch around opening, as indicated on pattern.	301	SSa-1	10-14	T-50	T-35	
	b. Tongue notch or die-cut opening and turn pocketing to inside. Top stitch 1/8 ± 1/16 inch from folded edge. The opening is to finish 3/4 +1/4, -1/16 inch wide.	301	SSa-1	10-14	T-50	T-35	
	c. Superimpose pocket facing shell on top sleeve as indicated on pattern. Insert atropine pocket flap, catching flap in stitching. Stitch around opening.	301	SSa-1	10-14	T-50	T-35	
	d. Tongue notch or die-cut pocket opening. Turn facing to inside and top stitch 1/8 ± 1/16 inch from all four edges.	301	SSa-1	10-14	T-50	T-35	
	e. Superimpose top sleeve pocket opening on top of lining opening. Stitch lower pocket opening together and ends of pocket opening 1/8 ± 1/16 inch from edge.	301	SSa-1	10-14	T-50	T-35	
	f. Close the top and sides of opening by folding down the outer shell sleeve and stitching through the facing and flap raw edges while penetrating sleeve liner and butyl pocketing.	301	SSa-1	10-14	T-50	T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
23.	Make atropine pocket and opening. (cont'd)						
	g. Fold butyl pocketing up so that the top and bottom edges are even, and stitch pocketing to liner 1/8 to 1/4 inch from edges on all four sides.	301	SSa-1	10-14	T-50	T-35	
	h. When this operation is complete the pocket depth shall be 6 ± 1/2 inches.						
	NOTE: When this operation is complete and the fastener tapes are engaged, the atropine pocket flap should not lay flat. (The blouse effect is to accommodate the atropine kit.)						
	or						
	i. Position butyl pocketing on sleeve liner (foam side) and stitch and around opening, as indicated on pattern.	301	SSa-1	10-14	50	50	
	j. Tongue notch or die cut opening and turn pocketing to inside. Topstitch 1/8 ± 1/16 inch from folded edge. The opening is to finish 3/4 +1/4, -1/16 inch wide.	301	SSa-1	10-14	50	50	
	k. Fold butyl pocketing up so that the top and bottom edges will be even and stitch pocketing to liner 1/8 to 1/4 inch from edge on all four sides.	301	SSa-1	10-14	50	50	
	l. Superimpose pocket facing shell on top sleeve as indicated on pattern. Insert pocket flap, catching flap in stitching and stitch on all four sides or insert pocket flap between facing and sleeve. Stitch top and bottom of pocket opening using two rows of stitching 3/4 inch apart.	301	SSa-1	10-14	50	50	

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					NEEDLE	BOBBIN/ LOOPER	COVER
	m. Tongue notch or die cut pocket opening Turn facing to inside and topstitch 1/8 ± 1/16 inch from edges catching facings while stitching on sides across bottom and 5/8 inch across top from both ends.	301	SSa-1	10-14	50	50	
	n. Superimpose top sleeve pocket opening on top of lining opening. Stitch lower pocket openings together across bottom opening and ends of pocket opening 1/8 ± 1/16 inch from edge.	301	SSa-1	10-14	50	50	
	o. Superimpose edge stitching on top of pocket opening and stitch 1/8 ± 1/16 through all plies.	301	SSa-1	10-14	50	50	
	p. When this operation is complete, the pocket depth shall finish 6 ± 1/2 inches.						
	NOTE: When this operation is complete, the fastener tapes are engaged, the atropine pocket flap should not lay flat. (The blouse effect is to accommodate the atropine kit.)						
24.	Set sleeves and join side seams.						
	a. Join sleeve lining to coat lining with a 1/2 ± 1/8 inch seam.	301 or 401	SSa-1	8-10	T-50 T-50	T-35 T-35	
	b. Join the sleeve shell to the coat shell with a double-lapped, double-stitched seam. The coat shell shall overlap the sleeve.	301 or 401	LSc-2	10-14	T-50 T-50	T-35 T-35	
	c. In one operation, join the lining under arm and side body seams with a 1/2 ± 1/8 inch seam.	301 or 401	SSa-1	8-10	T-50 T-50	T-35 T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
24.	Set sleeves and join side seams. (cont'd)						
	d. In one operation, join the outer shell under arm and outer shell side body seams with a double-lapped, double-stitched seam. The fronts shall overlap the backs.	301 or 401	LSc-2	10-14	T-50 T-50	T-35 T-35	
25.	Finish sleeve bottoms.						
	a. With the lining even with the finished hem of the outer shell, turn in bottom edge of outer shell as indicated on pattern. Fold in raw edges of outer shell 1/4 ± 1/16 inch, stitch the hem completely around the bottom of the sleeve.	301	SS1-1	10-14	T-50	T-35	
	b. The sleeve adjustment tab shall be assembled in the same manner as operations 12.a and 12.b.						
	c. The loop fastener tape and sleeve adjustment tab shall be stitched to outer shell sleeve as indicated on pattern.						
26.	Make curtain for coat.						
	a. Turn center front edges of curtain to inside, as indicated on pattern, and with raw edges turned under stitch 1/8 ± 1/16 inch from turned under edges. This facing should finish 1/2 ± 1/8 inch wide.	301	EFb-1	10-14	T-50	T-35	
	b. Turn up hem of curtain to inside, and stitch 3/8 ± 1/16 inch from hem edge. The hem shall finish 1/2 ± 1/8 inch wide.	301	EFb-1	10-14	T-50	T-35	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
27.	Make fastener tabs and attach to curtain.						
	a. Fold cotton tape in half over one end of elastic webbing, catching 1/4 inch of the braid between the cotton tape turned under 1/4 inch, stitch 1/8 ± 1/16 inch from all edges of tape.	301	OSf-1	10-14	T-50	T-35	
	b. Make buttonhole in cotton tape, as indicated on pattern.	Button-hole		52-56 per buttonhole	T-50	T-35	
28.	Attach curtain to liner.						
	a. Position raw edges of curtain and elastic tabs to knit side of liner as indicated on pattern.						
	b. Stitch curtain 1/8 inch from the raw edge while simultaneously catching the curtain reinforcement tape and elastic tabs. The reinforcement tape shall be positioned on the foam side of the liner and shall be held taut during stitching. Turn curtain down and raise stitch 3/16 to 1/4 inch from turned edge, burying the raw edges within the seam.	301	Similar to LSb1-2	10-14	T-50	T-35	
	NOTE: The curtain shall finish so that the PTFE laminate side faces the wearer.						
29.	Finish bottom of coat.						
	a. Make one buttonhole or sewn eyelet at the bottom right front coat as indicated on pattern in lieu of metal eyelets.	401		19-24 per eyelet	T-50	T-35	
		or 502		26-32 per eyelet	T-50	T-35	
		or 503		26-32 per eyelet	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
29.	Finish bottom of coat. (cont'd)						
	b. Position the elastic drawcord on inside of coat with ends exiting through eyelet and knot ends of drawcord after applying a spring loaded cord closure.						
	NOTE: On the left front, the drawcord will simply extend out from the casing formed by the hem.						
	c. Turn up bottom edge of outer shell even with the lining and with the raw edge turned in stitch 1/8 to 3/16 inch from turned edge with the stitching continuing along the complete bottom forming a 3/8 ± 1/16 inch pleat in the center bottom of the coat liner. The hem shall finish 1 ± 1/8 inches from the bottom edge of the coat.	301	SS1-1	10-14	T-50	T-35	
	d. Center the elastic drawcord in the hem of the coat. Put a bartack on the drawcord in the back center through the hem making sure to catch the drawcord.	Bartack		28 per bartack	T-50	T-35	
30.	Attach slide fastener.						
	a. Position the back edge of the slide fastener tape on the inside of the left front, 2-3/8 ± 1/8 inches from the finished front edge, as indicated on the pattern, with top stop 1/2 to 5/8 inch from the top.						
	b. Turn under top tape and stitch tape to coat 1/16 to 1/8 inch from back edge.	301	LSbj-1	10-14	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
30.	Attach slide fastener. (cont'd)						
	c. Place a second row of stitching through tape and coat positioned 1/4 ± 1/16 inch from the first row of stitching.	301	LSbj-1	10-14	T-50	T-35	
	d. Position the back edge of slide fastener tape (tape containing slider) on outside of right front 2-3/8 ± 1/8 inches from the finished front edge as indicated on patterns with the top stop 1/2 to 5/8 inch from neck edge, with the top end of tape turned under (+0, -1/4 inch tolerance).						
	e. Turn the tape under and stitch tape to coat 1/16 to 1/8 inch from back edge.	301	LSd-1 or LSbj-1	10-14	T-50	T-35	
	f. Place a second row of stitching through tape and coat positioned 1/4 ± 1/16 inch from the first row of stitching.	301	LSd-1 or LSbj-1	10-14	T-50	T-35	
	g. When using a double needle machine, turn the coat and continue stitching across the turned under ends of the tape (perpendicular to the slide fastener) and backtack to secure, forming a box stitch.						
	or						
	h. Bartack across the turned under ends of the tape (perpendicular to the slide fastener).	Bartack		28 per bartack	T-50	T-35	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
31.	Make cones and attach to coat.						
	a. Cut a round hole through outer shell and liner at the side seams, as indicated on pattern, and turn raw edges to the inside such that the hole finishes 2 inches in diameter. Stitch 1/8 to 1/16 inch around the edges.	301	EFa-1	10-14	T-50	T-35	
	b. Place two lengths of 1 inch wide flat elastic webbing (each cut 3 ± 1/8 inches in length), with lengthwise edges parallel, centered over the hole on the inside of the coat. Stitch to coat through all layers, leaving parallel edges over hole open to allow the passage of a hose.	301	LSbj-1	10-14	T-50	T-35	
	c. Join seams in both cone shell and lining pieces.	301	SSa-1	10-14	T-50	T-35	
	d. Slip cone lining into cone shell, with wrong sides together. Edge stitch raw edges of tape and bottom of the cone.	301 Class 500	SSa-1	10-14	T-50	T-35	
	e. Sew two eyelets at top of cone as indicated on pattern.	Eyelet		19-24 per eyelet	T-50	T-35	
	f. Position the elastic cord on the inside of the cone with ends exiting through eyelet. Draw both ends through one cord closure and knot each cord end.						
	g. Turn back top edge of cone 1/2 ± 1/8 inch enclosing the elastic cord; and stitch 1/16 to 1/8 inch from raw edges.	301 or 401	EFa-1	10-14	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
31.	Make cones and attach to coat. (cont'd)						
	h. Attach cone to coat, with layer end of cone centered over the hole on the outer shell, and the cord closure facing towards the front of the coat. Stitch through all layers of coat, 1/8 ± 1/16 inch around the edge taking care not to stitch over biswing.	301	LSbj-1	10-14	T-50	T-35	
	Make Trousers						
32.	Join trouser fronts to backs at sides.						
	Join the outer right front shell to outer right back shell with a double-lapped, double-stitched seam, the front to overlap the back. In similar fashion join the left front outer shell to the left back outer shell, the front to overlap the back.	301	LSc-2	10-14	T-50	T-35	
33.	Make and attach calf pockets and flaps.						
	a. The calf pockets and flaps will be constructed in the same manner as operations 20.a., b., and c. and 21.a., b., e., and f. In addition, the lower corner of the bellows pocket shall have a diagonal row of reinforcement stitching running from the pleat corner at a 45° angle into the pocket area for 2 ± 1/8 inches, backtacked 1/2 ± 1/8 inch at both ends.	301	LSd-1	10-14	T-50	T-35	
	b. Position the calf pocket protective piece as indicated on the pattern. Stitch to outer shell, 1/8 ± 1/16 inch around all the edges.	301	LSbj-1	10-14	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
33.	Make and attach calf pockets and flaps. (cont'd)						
	c. Position pocket on the trouser shell over the protective pieces and turning in the raw edges. Stitch along the edges with a 1/8 ± 1/16 inch raised stitch. The protective pieces shall not extend beyond the sides or bottom of the pockets. The bellows shall finish to the rear. Place a bartack at the top corner (bellows side).	301	LSd-1	10-14	T-50	T-35	
		Bartack		28 per bartack	T-50	T-35	
	d. Position raw edges of pocket flaps above the pocket opening, as indicated on patterns and stitch to trouser shell 1/8 to 3/16 inch from raw edges. Turn flaps down and raise stitch 3/16 to 1/4 inch from turned edge, burying raw edges within the seam. The protective pieces shall not extend beyond top and side edges of flaps.	301	LSb1-2	10-14	T-50	T-35	
34.	Make and attach thigh pocket.						
	a. Fold thigh pocket welt as indicated on pattern and position along center of slide fasteners, and stitch 1/2 inch from folded edge.	301 or 401	LSd-1	10-14	T-50	T-35	
	b. Stitch the cut-out corners of thigh pocket, face to face, forming the bellows. Turn seam toward the bottom of pocket and top stitch 1/16 to 1/8 inch from seam.	301 or 401	LSq-2	10-14	T-50	T-35	
	c. As indicated on pattern, fold both the side and bottom with the wrong sides together and raise stitch 1/16 to 1/8 inch from edges.	301	OSf-1	10-14	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
34.	Make and attach thigh pocket. (cont'd)						
	d. To form bellow pleat, fold bottom and sides as indicated on pattern, face to face, and raise stitch 1/16 to 1/8 inch from edges.	301	OSf-1	10-14	T-50	T-35	
	e. Turn under thigh pocket side (side not bellowed), as indicated on pattern and position along center of slide fastener, abutting thigh pocket welt. Stitch 1/2 inch from edge. NOTE: Slide fastener shall finish so that the bottom stop is located at thigh pocket top.	301 or 401	LSd-1	10-14	T-50	T-35	
	f. Position thigh protective piece as indicated on pattern. Stitch to outer shell 1/8 ± 1/16 inch around all the edges.	301 or 401	LSbj-1	10-14	T-50	T-35	
	g. Position thigh pocket assembly on thigh protective piece, with bellows toward the rear. Turn under raw edges and raise stitch 1/16 to 1/8 inch on all sides. The protective piece shall not extend beyond the sides or bottom of the thigh pocket.	301 or 401	LSd-1	10-14	T-50	T-35	
	h. Place a horizontal bartack at each end of the slide fastener, on inside of raised stitching, reinforcing the slide fastener opening.	Bartack		28 per bartack	T-50	T-35	
35.	Make and attach slide fastener cover for the ankle adjustment tab.						
	As indicated on pattern, join slide fastener cover, face to face, with 1/2 ± 1/16 inch seam for 2-3/8 ± 1/8 inches from top. Position slide fastener below seam and to inside of each opening with raw edges	301 or 401	LSb-1	10-14	T-50 T-50	T-35 T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
35.	Make and attach slide fastener cover for the ankle adjustment tab. (cont'd) turned under, and with top slide stop 1-1/8 ± 1/8 inches from open end of slide fastener cover. Stitch slide fastener to cover 3/8 ± 1/16 inch from opening with stitching extending around both sides and top forming a U-shape.						
36.	Attach slide fastener and cover to inseam (after inseams are joined) of outer shell trouser leg.						
	Place slide fastener and cover in an open position on the outer shell as indicated on pattern. Turn in edges and raise stitch 1/16 to 1/8 inch around the edges.	301 or 401	LSb-1	10-14	T-50 T-50	T-35 T-35	
	NOTE: When slide fastener is closed the leg opening will be narrower and the slider shall be at the bottom of the trouser leg.						
37.	Join crotch, seat and inseams of outer shell.						
	a. Position left front to right front at crotch and join the crotch seam beginning at the inseam extending 3-1/2 ± 1/8 inches to seam termination. Turn on right side and place a row of single stitching on each side of seams 1/8 inch away.	301 or 401	LSq-3	10-14	T-50	T-35	
	b. Join seat seam of outer shell with double-lapped and double-stitched seam. The seat seam may lap either direction.	401	LSc-2	10-14	T-50	T-35	
	c. Join inseams with a double-lapped, and double-stitched seam and the front will overlap the back.	301 or 401	LSc-2	10-14	T-50 T-50	T-35 T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
38.	Attach ankle adjustment tabs. Construct and attach ankle adjustment tabs and hook and loop fastener tape as in operation 12. The ankle adjustment tab will be stitched on the trouser back, just rearward of the slide fastener cover. The loop fastener will be stitched to the trouser front, beneath the pocket, as indicated on pattern. NOTE: The ankle adjustment tab and loop tape will be stitched to the shell only.						
39.	Make and attach hip adjustment straps.						
	a. Thread front hip adjustment strap over first bar (opposite pull tab of buckle) and fold strap in half lengthwise so that bar is located at the fold. Fold under two raw ends of strap 5/16 ± 1/16 inch. Position strap with long edge parallel to waist and with turned under edge 1-5/16 ± 1/16 inches from outseam on the front, and 3-5/8 ± 1/8 inches below finished waist edge. Bartack and turn under across the width of strap with a bartack.	Bartack		28 per bartack	T-50	T-35	
	b. Turn back one end of back hip adjustment strap 1 ± 1/8 inch and then turn under raw end 5/16 ± 1/16 inch and box stitch 1/16 to 1/8 inch from edge or bartack at end.	301 or Bartack		8-10 28 per bartack	T-50 T-50	T-35 T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
39.	Make and attach hip adjustment straps. (cont'd)						
	c. Thread unfinished end of back hip adjustment strap through pull tab end of buckle by bringing the strap up from underneath the buckle, through the pull tab and the first bar, over the first bar and down through the buckle again. Pull unfinished end of strap toward trouser back until turned under end is at the pull tab.						
40.	Begin attaching belt loops.						
	a. All sizes shall have seven belt loops. All belt loops shall finish 1-5/8 ± 1/8 inches in length (measurement taken between bartacks) on each pair of trousers positioned as follows:						
	(1) One centered on each trouser front piece.						
	(2) One adjacent to each outseam.						
	(3) One centered on each back piece.						
	(4) One adjacent to seat seam.						
	b. Turn under bottom raw edge to finished position and bartack 1/8 to 3/16 inch from folded edge to outer shell.	Bartack		28 per bartack	T-50	T-35	
41.	Make trouser lining.						
	Join side seams, crotch seam to notch, seat seam and inseams of trouser lining, stitch all seams with material face to face.	301 or 401	SSa-1	8-10	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
42.	Attach label. Sew identification, size, (combination instruction, identification and size label, when used) and instruction label 1/8 to 3/16 inch from the edge on all four sides to the knit side of left back of lining material (positioned as indicated below) with the top edges 3-1/8 ± 1/8 inches below top edge. (1) Size or combination label - centered (± 1/2 inch) on left back. (2) Identification label (when used) - positioned to left of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching. (3) Instruction label - positioned to the right of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching. NOTE: Final row of stitches on overlapped edges may be continued to top of lining.	301	I.Sbj-1	8-10	T-50	T-35	
43.	Join outer shell and lining at front opening. Insert lining into the outer shell, turn raw edges under 3/8 ± 1/16 inch on outer shell and lining between plies at front opening and stitch 1/16 to 1/8 inch from folded edge. The stitching shall continue to the top edge of the outer shell and shall terminate at the notch at base of fly.	301	SSc-1	10-14	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
44.	Attach slide fastener at waist.						
	a. Position slide fastener to inside of right front with back edge of chain extending beyond front edge 3/16 ± 1/16 inch. Stitch to front with two rows of stitching 3/16 ± 1/16 inch gage, with first row 1/16 to 1/8 inch from back edge of tape. The top stop shall finish 1-3/8 ± 1/8 inches from top edge of finished waist. The stitching shall continue to top end of fastener tape or the tape shall be turned under.	301	SSbd-2	10-14	T-50	T-35	
	b. Position slide fastener to inside of left front with edge of opened chain positioned as indicated on pattern and stitch through tape to front of trousers with two rows of stitching 3/16 ± 1/16 inch gage, with first row 1/16 to 1/8 inch from back edge of tape. The top end of tape may be turned under. The top stop shall finish 1-3/8 ± 1/8 inches from top.	301	LSbj-2	10-14	T-50	T-35	
45.	Finish waist.						
	a. Turn in top edge of outer shell 1-1/4 ± 1/8 inches even with the lining material and with the raw edges turned in 1/4 inch, stitch 1/16 to 1/8 inch from turned edge, with the stitching continued along ends of front opening. The top ends of slide fastener tapes may be caught in the waist stitching (when applicable). The waist hem shall finish 3/4 ± 1/8 inch.	301	SS1-1	10-14	T-50	T-35	
	b. Attach buttons on waist of trousers as indicated on patterns.	101 or 103		14-16 per button	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
46.	Finish attaching belt loops.						
	a. Turn under top raw edge of belt loop and position even with top edge of waist. Bartack to trousers 1/8 to 3/16 inch from top of folded edge.	Bartack		28 per bartack	T-50	T-35	
	or						
	b. Position raw edge of belt loop 1/4 inch from top edge of finished waist. Bartack to trousers 1/8 to 3/16 inch from edge of waist to finish 1-1/2 to 1-3/4 inches in length.						
47.	Make fly protective flap.						
	a. Join outer shell and lining piece of protective flap along side and bottom edges, face to face.	301 or	SSe-2(a)	10-14	T-50	T-35	
	Turn, work out edges and stitch 3/16 to 1/4 inch from turned edges with stitching continued to the top edge of outer shell material.	401	SSe-2(b) and LSd-1	10-14	T-50	T-35	
	b. Turn in top edge of outer shell even with the raw edge of lining and stitch 1/16 to 1/8 inch from turned edge with the stitching continued along side and top edges of hem. The hem shall finish 1/2 to 5/8 inch in width.	301	SS1-1	10-14	T-50	T-35	
48.	Attach the fly protective flap.						
	a. Position protective flap on inside of left front, as indicated on pattern, with the outer shell toward the outside and join to trousers, 3/16 to 1/4 inch from back and bottom edge of flap. The fly protective flap may be bartacked at crotch.	301 or	LSbj-1	10-14	T-50	T-35	
		Bartack		28 per bartack	T-50	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
48.	Attach the fly protective flap. (cont'd)						
	b. Place a vertical bartack at top back edge of flap through trousers. The bartack shall be positioned 1/4 to 5/16 inch from top and back edges of flap.	Bartack		28 per bartack	T-50	T-35	
	or						
	c. As an alternate the bartack may be placed even with the top edge of fly protective flap and 1/4 to 5/16 inch from the back of the flap.						
49.	Finish leg bottom.						
	Turn up bottom edge of outer shell 1 ± 1/8 inch, even with lining and with raw edges of hem, slide fastener tape and slide fastener cover turned in. Stitch 1/8 to 3/16 inch from turned edge around the entire hem.	301	SS1-1	10-14	T-50	T-35	
	NOTE: There must be a break in the stitching at either side of the slide fastener so as not to interfere with the functioning of the slide fastener.						
50.	Attach snap fasteners.						
	a. Secure a double stud fastener to right front at waist, or as an alternate secure style 2 or 2A stud-eyelet combination (on right outside of waist opening) and socket (on right side of waist opening) as indicated on pattern.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
50.	Attach snap fasteners. (cont'd) b. Secure a female portion of snap fastener to left front at waist, positioned to correspond with male at right front at waist. c. Secure a socket or as an alternate a stud (if alternate in 50.a is used) portion of snap fastener to fly protective flap positioned to correspond with the double stud fastener on the right front. The fly protective flap shall lay smooth and flat when the socket and stud are engaged.						
51.	Clean coat and trousers. a. Trim ends of threads to 1/4 inch maximum length. b. Remove all loose threads. c. Remove all size markings.						

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3.8 Finished measurements. The finished measurements for coat and trousers shall be as shown in table II.

TABLE II. Finished measurements (inches)

Size	Coat				Trousers	
	Cross Back 1/	1/2 chest 2/	Sleeve inseam 3/	Back length 4/	1/2 waist 5/	Leg inseam 6/
X-Small - X-Short	16	21	19-1/2	25	---	---
X-Small - Short	16	21	20-1/2	27	14-1/2	30-3/4
Small - X-Short	17	23	19-1/2	26	16-1/2	28-3/4
Small - Short	17	23	20-1/2	27-1/2	16-1/2	30-3/4
Small - Regular	17	23	21-1/2	29	16-1/2	32-3/4
Small - Long	17	23	22-1/2	30-1/2	16-1/2	34-3/4
Medium - X-Short	18	25	19-1/2	26-1/2	18-1/2	28-3/4
Medium - Short	18	25	20-1/2	28	18-1/2	30-3/4
Medium - Regular	18	25	21-1/2	29-1/2	18-1/2	32-3/4
Medium - Long	18	25	22-1/2	31	18-1/2	34-3/4
Large - Short	19	27	20-1/2	28-1/2	20-1/2	30-3/4
Large - Regular	19	27	21-1/2	30	20-1/2	32-3/4
Large - Long	19	27	22-1/2	31-1/2	20-1/2	34-3/4
X-Large - Short	--	--	---	---	22-1/2	30-3/4
X-Large - Regular	20	29	21-1/2	30-1/2	22-1/2	32-3/4
X-Large - Long	20	29	22-1/2	32	22-1/2	34-3/4
Tolerance	± 3/4	± 3/4	± 1/2	± 3/4	± 3/4	± 1/2

1/ Measure across width of back from armhole seam to armhole seam along yoke seam.

2/ Measure across chest from folded side edge to folded side edge at base of armholes.

3/ Measure from armpit seam along inseam to bottom edge of sleeve.

4/ Measure down center of back from collar joining seam to bottom edge of coat.

5/ Measure across waist from side edge to side edge with snap fastener at waist secured and fly closed.

6/ Measure from crotch seam along inseam to bottom edge of leg.

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3.9 Carbon tetrachloride absorption. When tested in accordance with 4.4.7, the carbon tetrachloride absorption of the lining material in the end item shall be 1.2 mg/cm^2 with no one individual test result less than 1.0 mg/cm^2.

3.10 Workmanship. The finished coat and trouser shall conform to the product quality established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Utmost care shall be taken during component and subassembly fabrication to ensure quality workmanship and safety of the service person using the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics in 4.4.4 through 4.4.7.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Toxicological agents resistance testing. Testing for conformance to initial toxicological agents resistance, shall be government acceptance tested in accordance with Section 4 of MIL-C-12189, except that after 25 consecutively produced rolls has successfully met the requirements, one out of every two rolls will be randomly sampled and tested. In the event of a failure under the reduced sampling, 100 percent sampling will be reinstated until an additional 25 consecutive rolls have successfully met the requirements.

4.4.2 In-process inspection.

4.4.2.1 Outer shell. In-process inspection shall be performed on the fabricated outer shell. The presence of any of the following conditions shall be cause for rejection of the outer shell on which it occurs.

a. Any hole, cut, tear, needle chew or burn.
b. Any repair not meeting the requirements of 3.6.5.b.
c. Any bubbling, tunneling, or delamination of tape sealing repaired seams.

4.4.2.2 Desiccant exposure to atmosphere. The contractor's procedure for ensuring that desiccant units are not exposed to the atmosphere exceeding 2 hours (see 5.1.1) from time of unsealing component packaging to sealing in unit packs shall be described in his/her quality control or inspection plan.

4.4.2.3 Lining. In-process inspection shall be performed on the cloth and foam sides of laminated lining material during the cutting operations, and fabrication of the lining. Any hole, cut, tear, needle chew or burn, or any repair not meeting the requirements of 3.6.5, 3.6.5.a, 3.6.5.c and 3.6.5.d shall be cause for rejecting the lining on which the condition occurs. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance (see 6.5.2).

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4.4.2.4 End item critical defect examination. Prior to performing the end item sampling examination required in 4.4.3, the coat or pair of trousers shall be 100 percent examined for the critical defects in table III. This 100 percent end item critical defect examination shall become a part of the contractor's inspection system of quality program. Any coat or trousers found to contain one or more critical defects shall be rejected.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of coats or pair of trousers. The sample unit shall be one coat or pair of trousers, as applicable. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total (major and minor combined) defects. The AQLs shall be applied separately to the coat and trousers and failure of either the coat or trousers shall be cause for rejection of the lot. The finding of one or more critical defects shall be cause for rejection of the lot.

TABLE III. End item visual defects

Examine	Defect	Classification Critical	Major	Minor
General:				
Material defects and damages	Applicable to lining laminated cloth, foam side:			
	a. Any cut, tear, needle chew, hole, or burn thru the the laminated cloth:			
	- 1/16 to 1 inch in length inclusive		101	
	- more than 1 inch in length	1		
	b. Any cut, tear, hole, or burn in knit base fabric:			
	- 1/4 to 1 inch in length inclusive			201
	- more than 1 inch in length		102	
	c. Any mend:			
	- more than 1-1/4 inches in longest direction		103	
	- more than two mends per garment		104	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Material defects and damages (cont'd)	Applicable to outer shell cloth:			
	a. Any hole, needle chew, cut, tear, or burn:			
	- less than 1/8 inch in length		105	
	- 1/8 inch or longer	2		
	b. Mends:			
	- Any mend up to 1/4 inch in diameter or length			202
	- any mend more than 1/4 inch in diameter or length		106	
	- more than two mends	3		
	c. Any defect such as a smash, multiple float, or slub			203
	d. Any misweave, area of poor dye penetration, dye streak, broken or missing yarn, thin place, or shade bar			204
	Applicable to butyl cloth:			
	- any hole, cut, tear, or mend		107	
	- heavily coated side not facing outward		108	
	Applicable to hook and loop fastener tape:			
	- any cut, hole, or tear		109	
	- any missing yarn		110	
	- any broken yarn			205
	- hook or loop flattened		111	
	- width not as specified			206
	- any spot, stain, or streak			207
	- color not as specified			208
Shaded parts	Any outside part shaded except those cut from ends			209
Cleanness	Spot or stain (grease, oil, ink):			
	- less than 1 inch in longest dimension		112	
	- 1 inch or greater in longest dimension	4		

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Cleanness (cont'd)	NOTE: Stains attributed to charcoal content of lining should not be scored.			
	Five or more thread ends in excess of 1/2 inch, not trimmed from coat or trousers			210
	Two or more shade or size tickets or loose threads not removed from coat or trousers			211
Components and assembly	Any component part or required operation omitted, unless otherwise specified		113	
	Any assembly or operation not properly performed, unless otherwise classified herein.			212
	Any component not as specified		114	
Cutting	Any component part not cut in accordance with directional lines indicated on pattern or not in accordance with specified requirements		115	
Seams and stitching	Accuracy of seaming:			
	a. Seams twisted, puckered, or pleated (unless otherwise classified herein)			213
	b. Part of garment caught in unrelated operation or stitching		116	
	c. Ends of seams or stitching produced with 301 stitch type, when not caught in other seams or stitching, backtacked less than 1/4 or more than 1/2 inch			214
	d. Stitching overlapping end(s) of thread breaks less than 1/2 or more than 3/4 inch			215

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Seams and stitching (cont'd)	e. Ends of a continuous line of stitching not overlapped, or overlapped less than 1/2 or more than 3/4 inch			216
	f. Various shades of thread used on outside of same coat or trousers			217
	Gage/margin of stitching:			
	a. Gage of double needle stitching more or less than specified		117	
	b. Margin of edge or raised stitching more or less than specified:			
	- up to 2 inches inclusive			218
	- more than 2 inches		118	
	Open seam:			
	a. Any on joining seam of laminate cloth except those listed in c. below:			
	- up to 1/4 inch in length inclusive			219
	- more than 1/4 but not greater than 3/4 inch in length inclusive		119	
	- more than 3/4 inch in length	5		
	b. Any seam except joining seam of laminate cloth and those listed in c. below:			
	- up to 1/2 inch in length inclusive			220
	- more than 1/2 inch in length		120	
	c. Any on coat bottom, hem outside collar, trouser waistband turnunder, trouser leg bottom hem, sleeve hem, pocket stitchings, slide fastener tape, or on fastener tape:			
	- 1/4 to 1/2 inch in length inclusive			221
	- more than 1/2 inch in length		121	
	NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches on joining seam constitute an open seam. On doublestitched seams, a seam is considered open when one or both sides of the seam is open.			

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Seams and stitching (cont'd)	Raw edges (outer shell and liner):			
	a. On outside:			
	- up to 1/2 inch inclusive			222
	- more than 1/2 inch		122	
	b. On inside:			
	- any raw edge more than 1/2 inch (where edge is required to be turned in)			223
	NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.			
	Runoffs:			
	a. Joining seams: When resulting in an open seam, use "open seam" classification.			
	b. Edge or raised stitching (when not resulting in any open seam):			
	- 1/2 to 1 inch inclusive			224
	- more than 1 inch		123	
	Seam and stitch type:			
	a. Seam or stitch type not as specified			225
	b. Looper thread of 401 stitch type finishing on outside of coat or trousers		124	
	Stitch tension:			
	a. Loose tension, up to 1 inch (excluding laminate lining cloth)			226
	b. Loose tension, more than 1 inch (excluding laminate lining cloth)		125	
	c. Loose tension, up to 1 inch on laminate lining cloth		126	
	d. Loose tension, more than 1 inch on laminate lining cloth	6		
	e. Loose tension, on edge of raised stitching			227
	f. Tight tension (stitches break when normal strain is applied to the seam or stitching)		127	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Seams and stitching (cont'd)	NOTE: Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.			
	Stitches per inch (to be scored only when the condition exists on major portion of seam)			
	a. Less than specified:			
	- one or two stitches			228
	- more than two stitches		128	
	b. One or more stitches more than specified			229
Heat sealed seams	Edge of seam tape less than 3/16 inch from seam allowance		129	
	Any seam tape not located as specified			230
	Any required stitching not covered by seam tape			231
	Any needle punctures that have not been repaired using heat sealing tape			232
Seam tape adhesion	Seam tape lifting off fabric		130	
	Visible scorching (heat degradation of fabric on the laminate) in excess of 3/16 inch in width or 1/2 inch in length at any location along a taped seam		131	
Repairs	More than two repairs on either the coat or trousers		132	
	More than two heat sealing repairs on either the coat or trousers		133	
Thongs	Thong missing		134	
	Thong not specified size			233
	Ends of thong fray when tugged at			234

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Pockets and flaps	Edges of pockets pleated or twisted in stitching			235
	Raw edges of pocket hems not turned in			236
	Pocket flap tight causing fullness, twisting or curling of pocket flap			237
	Pocket protective piece exposed beyond side or bottom edges of pocket or exposed beyond side or top edges of flap			238
	Bellows portion of pockets positioned toward front of pocket		135	
	Hook and loop portions of fastener tape out of alignment when flap is laying flat			
	- more than 1/4 inch horizontally			239
	- more than 1/4 inch vertically			240
	Pocket flap not completely covering pocket opening by 3/16 inch or more			241
	Pocket or flap not uniform in shape and size by more than 1/4 inch			242
	Pocket or flap out of alignment by more than 1/2 inch			243
	Pocket or flap out of alignment with each other by more than 1/4 inch			244
	Corner of flap caught in bartacks at end of bellows opening, restricting freedom of flap			245
Buttonholes or eyelets	Not as specified			246
	Not in specified position			247
Slide fastener	Any part of slide fastener assembly omitted, bent, broken, cracked, or otherwise defective, affecting function	7		
	Bottom closed end stop not attached as specified		136	
	Not specified type or size		137	
	Fastener tape cut or torn		138	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Slide fastener (cont'd)	Shade of tape not as specified			248
	Not functioning properly, failing to effect a secure closure or to open freely		139	
	Fails to interlock chain		140	
	NOTE: The slide fasteners shall be fully closed and opened to determine if the slide fasteners are operable and provide a secure closure.			
Snap fastener	Any omitted, mismated, bent or broken		141	
	Any fastener not functioning properly i.e. fails to snap closed, provide a secure closure, or open freely		142	
	NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.			
	Clinched excessively tight, cutting adjacent material		143	
	Any fastener not aligned with each other creating a twist or buckle when closed			249
	Clinched loosely, permitting any component to rotate freely but not to the degree that any component can be expected to become detached during use			250
	Clinched loosely to the degree that components can be expected to come detached during use		144	
	NOTE: Incomplete roll of end of button or eyelet barrel is evidence of improper and insecure clinching.			

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
General (cont'd):				
Snap fastener (cont'd)	Incorrect style	145		
	More than three splits in eyelet or button barrel			251
	Not located as specified			252
Labels	Size, instruction, identification, or combination label (when applicable) missing, incorrect or illegible			253
	Surveillance marking omitted, incorrect or illegible	146		
	Label not stitched on all four sides	147		
	Stitched through printed portion of label			254
Bartacks	Bartack stitching loose, incomplete, broken, not fully engaging stitched material or not as specified:			
	- involving one bartack			255
	- involving two or more bartacks	148		
	Bartacks not as specified			256
	Bartack not located as specified			257
	Bartack missing			258
Coat:				
Pockets and flaps	Chest pocket pull tape missing or not properly caught in stitching			259
	Sleeve pocket not parallel with atropine pocket by 3/8 inch or more			260
	Cut lengths of tape for sleeve pockets and flaps less than 1-3/8 or more than 1-5/8 inches			261
	Cut lengths of tape for chest pockets less than 6-3/8 or more than 6-5/8 inches			262
	Cut lengths of hook and loop tape for sleeve adjustment tabs less than 4-3/8 or more than 4-5/8 inches			263

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Coat (cont'd):				
Pockets and flaps (cont'd)	Cut lengths of hook tape for sleeve adjustment tabs less than 3-3/8 or more than 3-5/8 inches			264
	Cut lengths of tape for center front overlap less than 1-7/8 or more than 2-1/8 inches			265
	Cut length of pull tape for chest pockets less than 4-3/4 or more than 5-1/4 inches			266
	Chest pockets out of parallel with respect to finished center front edge by more than 1/4 inch			267
	Chest pockets out of alignment between corresponding points by more than 1/2 inch (determination to be made by measuring from bottom of coat)			268
	Pencil carrier not as specified			269
	Pencil carrier not in specified location			270
Name tag	Not located above left chest pocket as specified			271
	Cut length of hook and loop tape less than 3-3/4 or more than 4-1/4 inches			272
Spring-loaded cord closures	Not as specified			273
	Not type or size specified			274
	Omitted, damaged or not functional	8		
Slide fastener	Top ends of tape not turned in and caught in the stitching joining slide fastener to coat			275
	Top stop less than 1/2 or more than 5/8 inch from neck edge, or bottom edge of tape more than 1/4 inch from bottom of coat or extending beyond bottom edge			276
	Back edge of slide fastener tape less than 2-1/4 or more than 2-1/2 inches from front edge			277

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Coat (cont'd):				
Drawcords	Dipping or impregnation omitted on any end			278
	Knot omitted on one or more cords			279
	Dipping or impregnation less than 1/2 inch in length			280
	Drawcord not extending from either front edges or cone			281
Collar	Edges of collar ends out of alignment by more than 1/4 inch (place shoulder seams together and extend collar ends to determine alignment)			282
Sleeves	Top sleeve not lapping under sleeve			283
	Ends of underarm seam and side seam staggered more than 1/2 inch (measured from center to center) of double rows of stitching			284
	Sleeve hem less than 3/8 or more than 5/8 inch in width			285
	Sleeve tab set on crookedly or poorly shaped			286
	Sleeve tab and loop tape out of alignment (measured at point of attachment to finished hem bottom in parallel direction):			
	- more than 3/8 and less than 3/4 inch			287
	- 3/4 inch or more		149	
	Sleeve tab and loop tape not able to be fastened	9		
	Width of sleeve tab less than 1 inch and more than 1-3/16 inches			288
Shoulder and armhole seams of outer shell	Fronts and back not lapping sleeves at armhole			289
	Fronts not lapping back at shoulder			290
Curtain	PTFE laminate side of finished curtain not facing wearer	10		
	Curtain reinforcement tape not located as specified			291

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Coat (cont'd):				
Curtain (cont'd)	Curtain reinforcement tape not as specified			292
	Curtain hem less than 3/8 or more than 5/8 inches in width			293
	Center front edges less than 3/8 or more than 5/8 inch in width			294
	Curtain not attached as specified			295
	Elastic tabs not as specified			296
Cone	Elastic cord caught in stitching			297
	Cone not positioned and attached as specified			298
	Hem less than 3/8 or more than 5/8 inch in width			299
	Elastic cord less than 11-1/2 or more than 12-1/2 inches in length			201a
	Knot omitted on ends			202a
Coat inner shell	Knit side of laminate material not finishing toward the inside	11		
	Seam allowance of joining seams not finishing toward the outer shell		150	
Waist adjustment tabs	Waist adjustment tabs set crookedly or poorly shaped			203a
	Waist adjustment tabs and loop tape out of alignment or not able to be fastened properly			204a
	Cut lengths of loop tape less than 4-3/8 or more than 4-5/8 inches			205a
	Cut lengths of hook tape less than 4-7/8 or more than 5-1/8			206a
	Not positioned as specified			207a
Bottom hem	Elastic drawcord caught in the stitching			208a
	Hem less than 1-1/8 or more than 1-3/8 inches in width			209a
	Twisted			210a
	Irregular in width by 1/8 inch or more			211a

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Trousers:				
Labels	Labels not positioned and attached to the knit side of right front lining as specified			212a
Leg slide fastener cover	Leg slide fastener cover stitched through lining material		151	
	Edges of leg slide fastener cover not abutted or overlapping 1/16 inch or more when fastener is closed			213a
	Ends of slide fastener tape not caught in the line of stitching at top of leg opening			214a
	With slide fastener in closed position, slider at top of opening	12		
	Thong omitted on trousers slide fastener pull		152	
	Slide fastener top stop (at trouser hem) located more than 1-1/4 inches from finished hem			215a
Belt loops	Less than 3/8 or more than 1/2 inch in width			216a
	Less than 1-1/2 or more than 1-3/4 inches in length between bartacks			217a
	Front edges of belt loops on fronts less than 2-1/2 or more than 2-3/4 inches from respective finished edges of front opening			218a
	One or both side of belt loops not on front or more than 1-1/4 inches from outseam			219a
	Back belt loops more than 1/2 inch off center between outseam and seat seam			220a
	Center back belt loop not positioned adjacent to seat seam			221a
	Set with the covering stitching on the outside			222a
	Top folded edge extending above or below top edge of waist more than 1/8 inch			223a

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Trousers (cont'd):				
Belt loops (cont'd)	One missing			224a
	Two or more missing		153	
Pockets and flaps	Cut lengths of calf pocket tape less than 1-7/8 or more than 2-1/8 inches			225a
	Side of pocket flap more than 1/4 inch from side of pocket edge			226a
	Thigh pocket slide fastener length not as specified			227a
	Slide fastener bottom stop not located at top of thigh pocket			228a
Hip adjustment strap and buckle	Improperly threaded, i.e., causing buckle to finish on reverse side or to function improperly		154	
	Top end of strap positioned more than 7/8 or less than 5/8 inch from top of finished waist			229a
	Turned under edge of front hip adjustment strap less than 1-1/4 or more than 1-3/8 inches forward of the outseam			230a
	Hem on free end of hip adjustment strap less than 5/8 or more than 3/4 inch or not boxstitched or bartacked as specified			231a
	Buckle defective or burred, affecting function		155	
	Hip adjustment straps (when loosened) not extending full length on back of trousers, or extending beyond their length, i.e., not permitting full expansion of top of trousers			232a
Front opening	Top stop on fastener tape less than 1-1/4 or more than 1-1/2 inches from finished top edge of waist			233a
	Back edge of slide fastener chain extending less than 1/8 or more than 1/4 inch beyond right front edge or opening			234a

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Trousers (cont'd):				
Front opening (cont'd)	Top edge of slide fastener tape not turned under when positioned on outside of right front			235a
	Double stud snap fastener not secured to right front		156	
	Thong omitted on trouser slide fastener pull		157	
Finished waist	Top hem turned down less than 5/8 or more than 7/8 inch			236a
Outseams, inseams, and crotch seam	Fronts not lapping back			237a
	Left front not lapping right front		158	
	Crotch and seat seam staggered more than 1/4 inch			238a
Leg hems	Bottom hem turned up less than 7/8 or more than 1-1/8 inches			239a
	Finished bottom hem less than 5/8 or more than 7/8 inch in width			240a
	Twisted			241a
	Irregular in width by 1/8 inch or more			242a
Ankle adjustment tabs	Ankle adjustment tab set on crookedly or poorly shaped			243a
	Ankle adjustment tab and loop tape out of alignment or not able to be fastened properly			244a
	Cut lengths of ankle adjustment tab hook tape less than 4-3/8 or more than 4-5/8 inches			245a
	Cut lengths of ankle closure loop tape less than 5-3/8 or more than 5-5/8 inches			246a

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Trousers (cont'd):				
Crotch seam (joining outer shell to lining)	Stitching joining outer shell to lining extending less than 1-1/4 or more than 1-1/2 inches below fly opening, or extending less than 3/4 or more than 1 inch above fly opening			247a
	Stitching joining outer shell to lining extending less than 3/16 or more than 5/16 inch across fly, or not superimposed on inside row of crotch joining seam			248a
Trousers lining	Knit side of laminate material not finishing toward the inside	13		
	Seam allowance of joining seams not finishing toward the outer shell		159	
Buttons	Not located or attached as specified			249a
	Not type, class or size specified			250a
	Omitted, damaged or not functional		160	
Labels	Label not positioned and attached to the knit side of back lining, as specified			251a

4.4.4 End item dimensional examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of coats or pair of trousers. The sample unit shall be one coat or pair of trousers, as applicable. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

DEFECTS

Any measurements deviating from any dimension specified in table II.

Sleeve lengths uneven by more than 1/2 inch

Leg lengths uneven by more than 1/2 inch

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4.4.5 Preproduction leakage test. Prior to seam sealing any coats or trousers, the basic PTFE laminated outer shell material shall be evaluated to assure proper seam sealing machine settings, tape alignment, appearance, and sealability. An individual evaluation shall be conducted per seam sealing machine at the beginning of each day upon which either coat or trouser end items are to be seam sealed. The following constitutes this evaluation:

a. Seam sealing procedure. Cut a 8 by 12 inch (any direction) sample of PTFE laminated outer shell material down the middle of the 8 inch material width for the entire 12 inch length. Double needle sew the cut edges together using a 301 or 401 stitch type, LSc-2 seam type at 10-14 stitches per inch with threads as specified in either 3.3.14.1 or 3.3.14.2. Operate seam sealing machine according to tape manufacturers recommended settings and apply middle of seam sealing tape over sewn seam on the PTFE (white) side of the material leaving at least a 1 inch peel tab of unsealed tape while still on the material backside. Observe PTFE material along the tape sides. There shall be no sign of melting, burning, delamination, puckering, discoloration, or any visible degradation of any kind to the PTFE material. Note that there may be slight roller marks due to the compressed rollers. These marks will exhibit a slightly darkened straight line effect which are permissible. Further, the tape shall be flat with no wrinkles, puckers, or threads protruding through its surfaces. Slowly pull three inches of the unsealed peel tab section of the tape. The white PTFE film laminate on the outer shell material shall evenly be delaminated across the full width of tape when peeled back.

b. Tape alignment. At random, fold a section of seam sealed material across the width of tape such that a crease mark is visible when unfolded. Observe that the tape edges remain secure with no sign of delamination at this point. Note that any lifted tape areas signify either delamination or points in which the heat nozzle of the sealing machine are in misalignment with the tape.

c. Hydrostatic resistance. To determine proper seal, a randomly selected taped section shall withstand a hydrostatic pressure of 50 cm for 3 minutes with no sign of moisture penetration through the seam or tape edges when tested in accordance with method 5516 of FED-STD-191. As an alternate, a commercially available hydrostatic machine with a 4-1/2 inch head may be utilized. The taped specimen shall maintain a hydrostatic pressure of 0.8 psi for 3 minutes with no sign of moisture penetration. In either case, the water pressure shall be applied to the side of the seam opposite the seam tape. In case of dispute, method 5516 shall be used.

d. Reporting. If previous taped material observation, tape alignment, and hydrostatic resistance results meet requirements, then record sealing machine settings (temperature, pressure and speed) and date onto taped material specimen with indelible ink, and continue with daily production utilizing these settings. Repeat all tests and reporting procedures prior to each day of production. Retain all taped specimens until acceptance of end items has been finalized. Also, the government reserves the right to repeat the preproduction leakage test anytime during a given day to verify machine settings.

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e. Failure. If any portion of the taped material, observation, tape alignment, and hydrostatic resistance evaluation fails, then production shall not begin until requirements have been met.

4.4.6 In-process hydrostatic testing. A sample unit shall be selected from each 100 units of coats or trousers in-process to be tested for hydrostatic resistance in accordance with 4.4.5.c. Selection shall be made after seam sealing, but prior to sewing in lining. Testing shall be accomplished without cutting or damaging the sample. Seams to be tested shall be as follows:

Coat	- Junction of shoulder and sleeve Underarm crossover Shoulder back seam Underarm straight seam
Trousers	- Pocket seam Seat seam One inseam One side seam

If there is evidence of leakage at any location on the sample unit, two additional randomly selected sample units shall be tested. If there is no evidence of leakage at any location on the two additional units, the 100 unit lot shall be accepted. Any evidence of leakage at any location on either of the two additional units shall be cause for rejecting the 100 unit lot.

4.4.7 End item testing. Samples of the coat or trousers shall be submitted to the Government (see 6.9) for acceptance testing for the characteristic specified in table IV. The lot size shall be expressed in units of coats or trousers, as applicable. The sample unit shall be one coat or trousers, as applicable. Any test failure shall be cause for rejection of the lot. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

TABLE IV. End item testing

Characteristic	Requirement paragraph	Test method	No. Determinations per sample unit	Results reported numerically to nearest
Carbon tetrachloride absorption	3.9	1/	2 2/	0.1 mg/cm^2

1/ The test procedure shall be as specified in MIL-C-43858, except that the test samples shall be conditioned in an air circulating oven at 50°C for at least 3 hours.

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2/ A sample of the seamless impregnated material shall be taken from an area adjacent to each of the following areas of the coat and trouser:

Coat
a. At the underarm including sleeve, side and armhole seams.
b. At the top of shoulder including shoulder and armhole seams.

Trousers
a. At the crotch including crotch, seat, and inseams.
b. At the side seam midway between the top of waist and top of thigh pocket.

4.4.7.1 Carbon tetrachloride warning. Carbon tetrachloride is a suspect carcinogen per OSHA and the American Conference of Governmental Industrial Hygienists - Threshold Limit Value (ACGIH-TLV) Booklet (ACGIH, Cincinnati, OH 1986, page 40). Personnel exposed to carbon tetracholoride should be informed of the potential hazards. Use should be contained in a ventilation hood. Ensure personnel use measures to prevent respiratory, skin, or other exposure by using personnel protective equipment. Consult the Material Safety Data Sheet (MSDS) for handling procedures and appropriate protective equipment.

4.4.8 Packaging examination.

4.4.8.1 In-process test of heat seal closure seam. Test shall be made daily prior to start of filling and sealing the unit packs to determine that heat sealing equipment is producing seals in accordance with the requirements of 5.1.1.1. One full width heat seal closure seam shall be made on an empty barrier bag using recommended heat seal settings and tested as specified in 4.5.2. Any test failure shall be cause for readjustment of the sealer settings and retesting until a satisfactory seam is produced.

4.4.8.2 In-process unit pack examination. Every filled and sealed unit pack shall be examined for the defects listed below. The finding of any defect shall be cause for rejection of the unit pack. Coats and trousers from rejected unit packs may be re-unit packed as specified in 5.1.1.1 and submitted for re-examination.

Defect

Closure seal width not as specified.
Closure seal not continuous.
Tear, cut, or hole in unit pack.
Specified number of desiccant units not enclosed.

4.4.8.3 Unit pack leakage testing. Every filled and sealed unit pack, prior to overpacking as specified in 5.1.1.2, shall be tested as specified in 4.5.1 for conformance to the leakage requirements in 5.1.1.1. Any unit pack which fails the test in 4.5.1 shall be opened and the suit shall be removed, thoroughly dried, re-unit packed as specified in 5.1.1.1, and retested as above.

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4.4.8.4 Overpack volume testing. Filled and sealed overpacks shall be tested as specified in 4.5.3 for conformance to the volume requirements in 5.1.1.1. The lot size shall be expressed in overpacks. The sample unit shall be one overpack. The inspection level shall be S-2. Any failure shall result in a rejection of the lot.

4.4.8.5 Overpack examination. Every filled and sealed overpack shall be examined for the defects listed below. The finding of any defect shall be cause for rejection of the overpack. Unit packs from rejected overpacks may be re-overpacked as specified in 5.1.1.2 and submitted for reexamination.

Defect
Closure seal width not as specified
Closure seal not continuous
Tear, cut, or hole in overpack

4.4.8.6 Shipping container examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or incomplete waterproofing. Bulged or distorted container. Inside dimensions of container not as specified. Partition not as specified.
Content	Number of unit packs per container less than required. Size shown on one or more unit packs not as specified on container. 1/

1/ For this defect, one unit pack from each container in the sample shall be examined.

4.4.8.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Leakage test. Each unit pack shall be tested for leaks in accordance with the hot water technique of Method 5009 of FED-STD-101. All samples shall be conditioned at ambient conditions for at least 4 hours prior to performing the hot water test. The time that the item and all processing materials have been maintained at ambient conditions prior to or during the processing period may be considered a part of the conditioning time. Bubbles which may appear on the surface of the unit pack but are not released at a slowly decreasing rate are not to be construed as an indication of failure. A steady stream of recurring succession of bubbles from any surface or seam shall be considered a failure. Upon completion of the test, the unit packs shall be thoroughly dried prior to overpacking.

4.5.2 Sealed seam test. Test shall be performed in accordance with Method 2024 of FED-STD-101 at normal room temperature with a static load of 50 ounces using fabricated bag specimen procedure. A 5-percent reduction in static load is permitted when room temperature in the test area exceeds 90°F.

4.5.3 Volume test. The total volume of the overpack shall be determined by a water displacement method. The measuring container shall have a surface area of not more than 300 square inches. The container shall be filled with water to a level that allows total submersion of the overpack. The temperature of the water shall be maintained between 110° and 140°F. A vertical gauge shall be mounted on the container so that the change in water level can be determined. Each overpack shall be placed in water and held under the water so that the top surface of the bag is 1/8 to 2 inches below the surface of the water. When the water turbulence has subsided to less than 1/8 inch total fluctuation, the vertical gauge shall be read. Calculate the total displacement of water to determine volume (see 5.1.1.1).

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation.

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5.1.1 Unit packing. One coat and two desiccant units or one pair of trousers and two desiccant units shall be folded so that the desiccant and all grommets, buckles and slide fasteners are not exposed. Desiccants shall conform to type II of MIL-D-3464. The folded coat or trousers shall measure approximately 14 inches in length, 12 inches in width and 4 inches in depth. The coat or trousers, together with a use and care instruction book, shall be unit packed in a heat sealable, flexible, water vaporproof, greaseproof barrier bag conforming to type I, class E, style 1 of MIL-B-117. The total time of exposure of the desiccant to the atmosphere after removal from its sealed container until final colsure of the barrier bag shall not exceed 2 hours. The outer surfaces of the barrier bag shall be uniformly colored overall with a color in the range of 34079 to 34087, inclusive, of FED-STD-595. The barrier bag shall be flat style, having inside dimensions of 20 by 15 $\pm$ 1/4 inches. As an option, to make a neat unit pack and to meet size requirements, the corners of the barrier bag may be heat sealed on a 45 degree angle approximately 3 inches along each side of the bag and trimmed. Excess air in the bag shall be removed by physical compression or by use of a partial vacuum (not to exceed 15 inches of mercury). The closure shall be heat sealed. The final seal shall be straight, continuous, parallel, and close to the edge of the bag. Closure seals shall show no separation when tested in accordance with 4.4.8.1. The filled unit pack shall not leak when tested in accordance with 4.4.8.3.

5.1.1.2 Overpacking. Each coat or pair of trousers, unit packed as specified in 5.1.1.1, shall be thoroughly dried after testing as specified in 4.4.8.3 and overpacked in a preformed heat sealed pouch conforming to size 2 of MIL-P-44406. The corners of the unit packed coat and trousers shall be folded onto itself, as necessary and in such a manner as not to puncture either the pouch material or the coat and trousers unit pack material and to facilitate the heat seal closing of the pouch. The large opening of the pouch shall than be closed using a continuous heat seal formed with a heat or impulse sealer. The heat seal shall be 3/16 to 1/4 inch in width, uniform and smooth (without wrinkles or foreign matter) and shall have a breaking strength of not less than 10 pounds per inch of width when tested in accordance with ASTM D 882 with the heat seal perpendicular to and centered in the long dimension of the test specimens. A vacuum tube shall be inserted into the pouch through the vacuum port. Excess air trapped between the unit pack and the pouch shall be extracted using a vacuum. After removal of the excess air, the vacuum port of the pouch shall be pinched-off as the vacuum tube is being withdrawn and then immediately closed with a heat seal. The overpack shall have a total volume of not more than 475 cubic inches, when tested in accordance with 4.4.8.4.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2).

5.2.1 Level A packing. Six coats or six pairs of trousers of one size and one lot only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Inside dimensionsof each shipping container shall be 24 inches in length, 14 inches in width, and 12 inches in depth. Each container shall be closed in accordance with method III, waterproofed, and reinforced as specified in the appendix of

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PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Six coats or six pairs of trousers of one size and one lot only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V3c of PPP-B-636. Inside dimensions of each shipping container shall be 24 inches in length, 14 inches in width and 12 inches in depth. Each container shall be closed in accordance with method III, and waterproofed in accordance with method V as specified in the appendix of PPP-B-636.

5.3 Palletization. When specified (see 6.2), unit packs packed as specified in 5.2.2 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129. Inspection/test data as specified in MIL-STD-129 for shelf life markings shall be based on the date of impregnation of the foam material and not the date that the unit is manufactured. (The last six digits of the surveillance number are the day, month and year of impregnation.) Additionally, unit packs shall be marked as specified in 5.4.1 and 5.4.2. All unit pack markings, except for the fit and size prediction legend, shall be marked on the same face of the vapor barrier bag. The fit and size prediction lengend shall be marked on the reverse face.

5.4.1 Surveillance marking.

5.4.1.1 Unit pack. Each unit pack shall be marked with the surveillance marking indicated on the enclosed coat and trouser (see 3.3.15.1). The markings shall be printed or stamped below the identification and contract data markings in 1/2 inch high black characters.

5.4.1.2 Shipping container. Each shipping container shall be marked with the surveillance marking indicated on the enclosed coat and trouser (see 3.3.15.1). The markings shall be stenciled, stamped or printed on the same side of the container as the identification and contract data markings and one end of the container in 1 inch high black characters.

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5.4.2 Fit and size prediction legend markings. Each unit pack shall be marked with a fit and size prediction legend as specified in 3.3.15.5. Markings shall be printed in black, bold face characters. The item identification characters shall be a minimum of 18 point (approximately 1/4 inch in height) and the instructive information characters shall be a minimum of 10 point (approximately 1/8 inch in height).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coat and trousers are intended for use by aviator personnel to provide protection against battlefield concentrations of chemical agents in liquid, vapor, or aerosol form.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Size(s) required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
e. When a first article sample is required (see 3.1, 4.3, and 6.3).
f. Level of packing (see 5.2).
g. Type and class of unit load required (see 5.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample and shade sample. For access to sample of the end item and samples for shade, address the contracting activity issuing the invitation for bids.

6.5 Warning.

6.5.1 Protection of processed goods. Processed goods should be protected from exposure to chemical vapors such as solvents to prevent contamination of the activated carbon.

6.5.2 Personnel protection. During the handling of the charcoal containing material used in fabricating the Coat and Trousers, Chemical Protective, Air Crew, Flame Resistant, charcoal from the material may collect on various

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surfaces; for example, on the skin side of the personnel, particularly on the hands, and on sewing machines and cutting tables. Therefore, it is essential that the industrial hygiene procedures required by the Occupational Safety and Health Standards be closely adhered to during the manufacturing of the suit. Provide general and local exhaust ventilation sufficient to maintain airborne dust concentration below that listed in Title 29, CFR, Part 1910.1000 OSHA. Otherwise respiratory protection may be required due to charcoal dust generation. Personnel should wear gloves and protective clothing when handling the material. Chemical safety goggles should be worn whenever eye contact may occur. Personnel should not wear contact lenses. Also, washing facilities and eyewash stations should be available in the work areas.

6.6 Water repellent treatment, MIL-T-44100. Either approval from the U.S. Army Natick Research, Development, and Engineering Center or a certificate of compliance stating that the aramid PTFE finish conforms to MIL-T-3530 vertical rise wicking requirements will be acceptable.

6.7 Suggested source for cord closure. A suggested source for the cord closure conforming to ITW Nexus Barreloc (see 3.3.10) is ITW Nexus, 230 West Gerry Drive, Wooddale, Illinois 60191, telephone number (312) 595-1888.

6.8 Equal item. Prior to the use of an "or equal" item, the supplier shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.9 Address for toxicological agents testing (see 4.4.7). The address to which the sample of coats or trousers for resistance to toxicological agents testing should be forwarded to:

Commander
Defense Personnel Support Center
2800 South 20th Street
Philadelphia, PA 19101

6.10 Subject term (key word) listing.

Aviator
Crewchief
Duty uniform

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Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

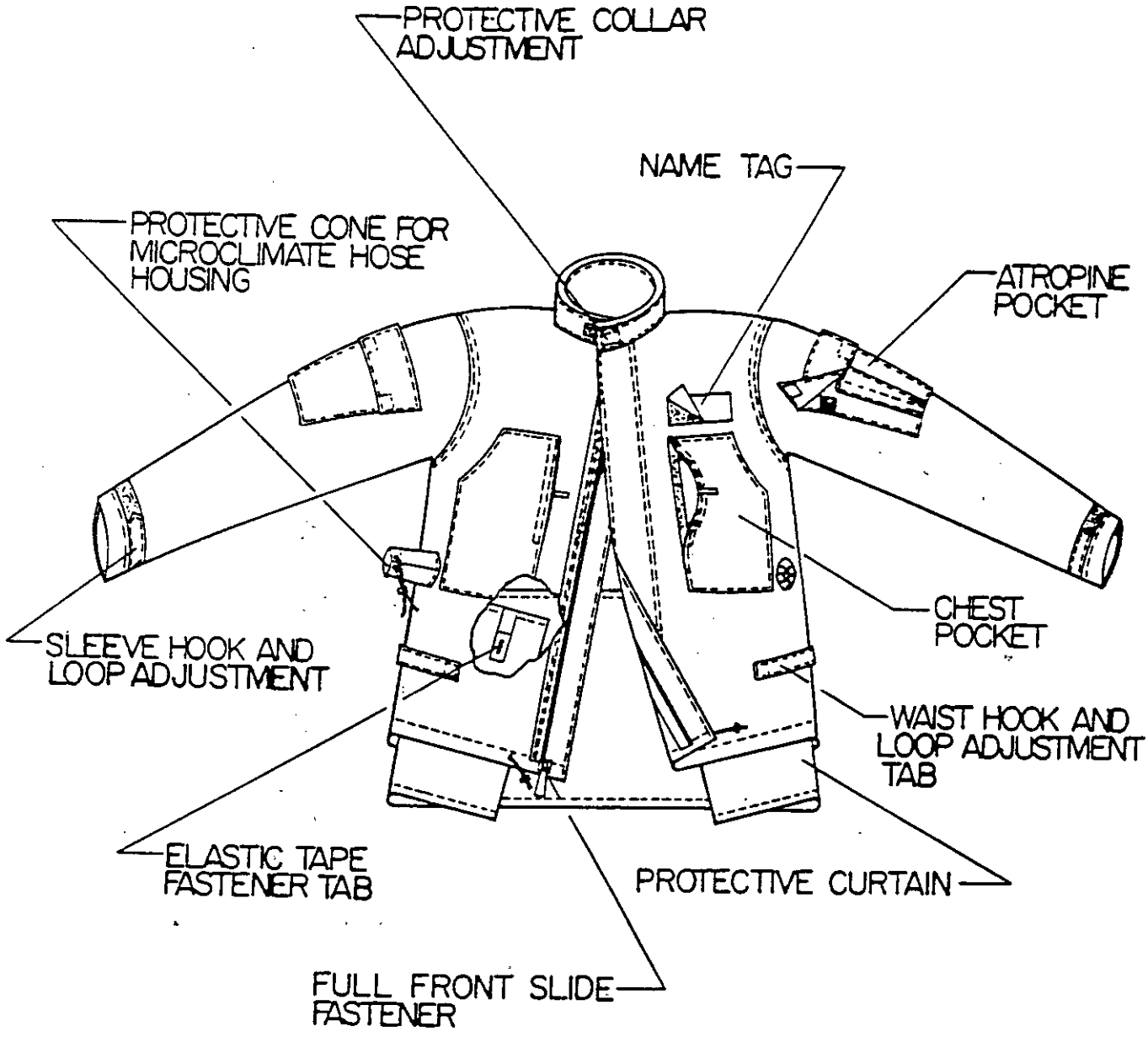
Army - MD, EA
Navy - SH, MC
Air Force - 99
DLA - CT

Preparing activity:

Army - GL

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FRONT VIEW

FIG 1, COAT, CHEMICAL PROTECTIVE: AIR CREW, FLAME RESISTANT

2-1-1855

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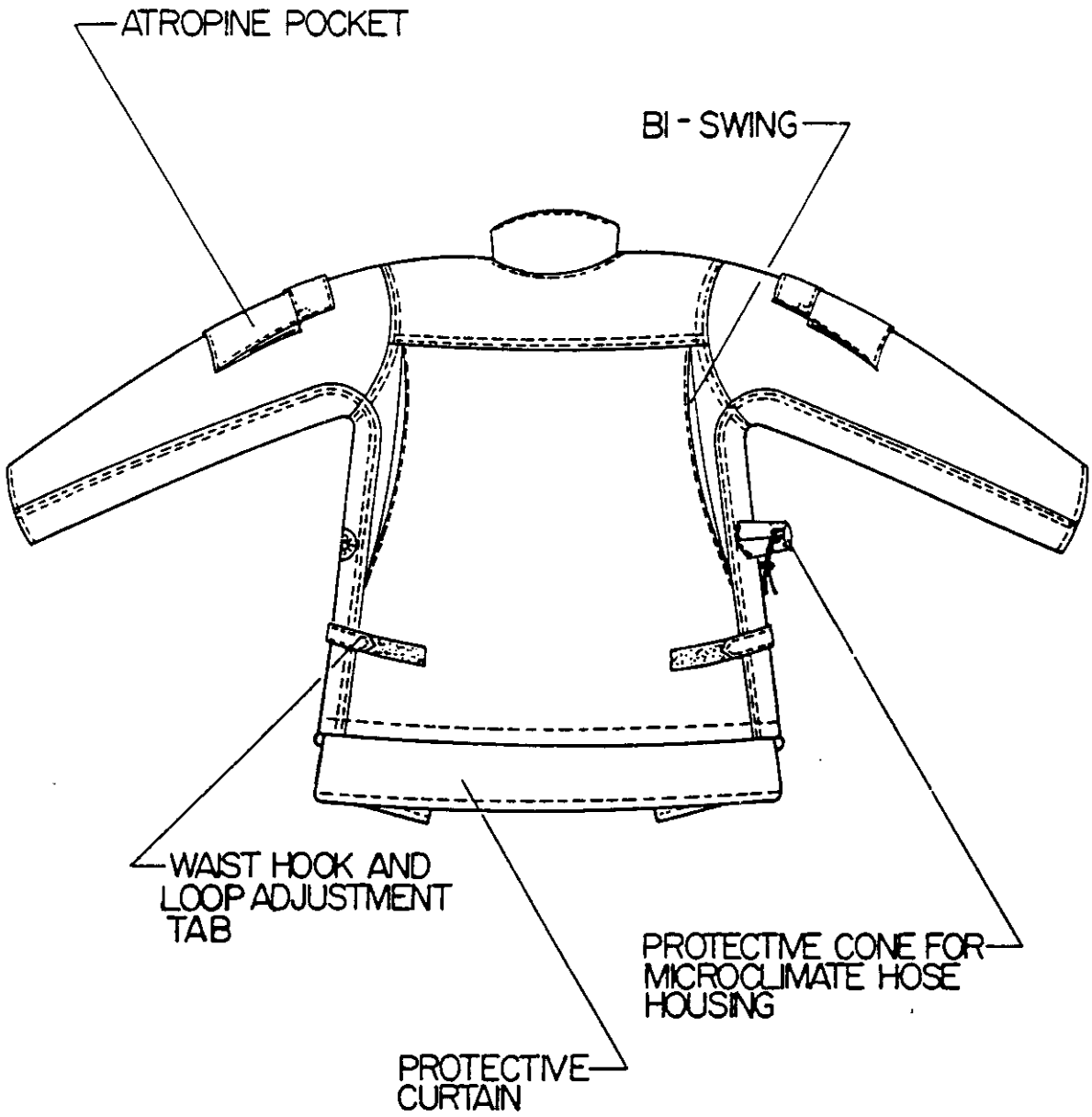


FIG. 2, COAT, CHEMICAL PROTECTIVE: AIR CREW, FLAME

2-1-1855

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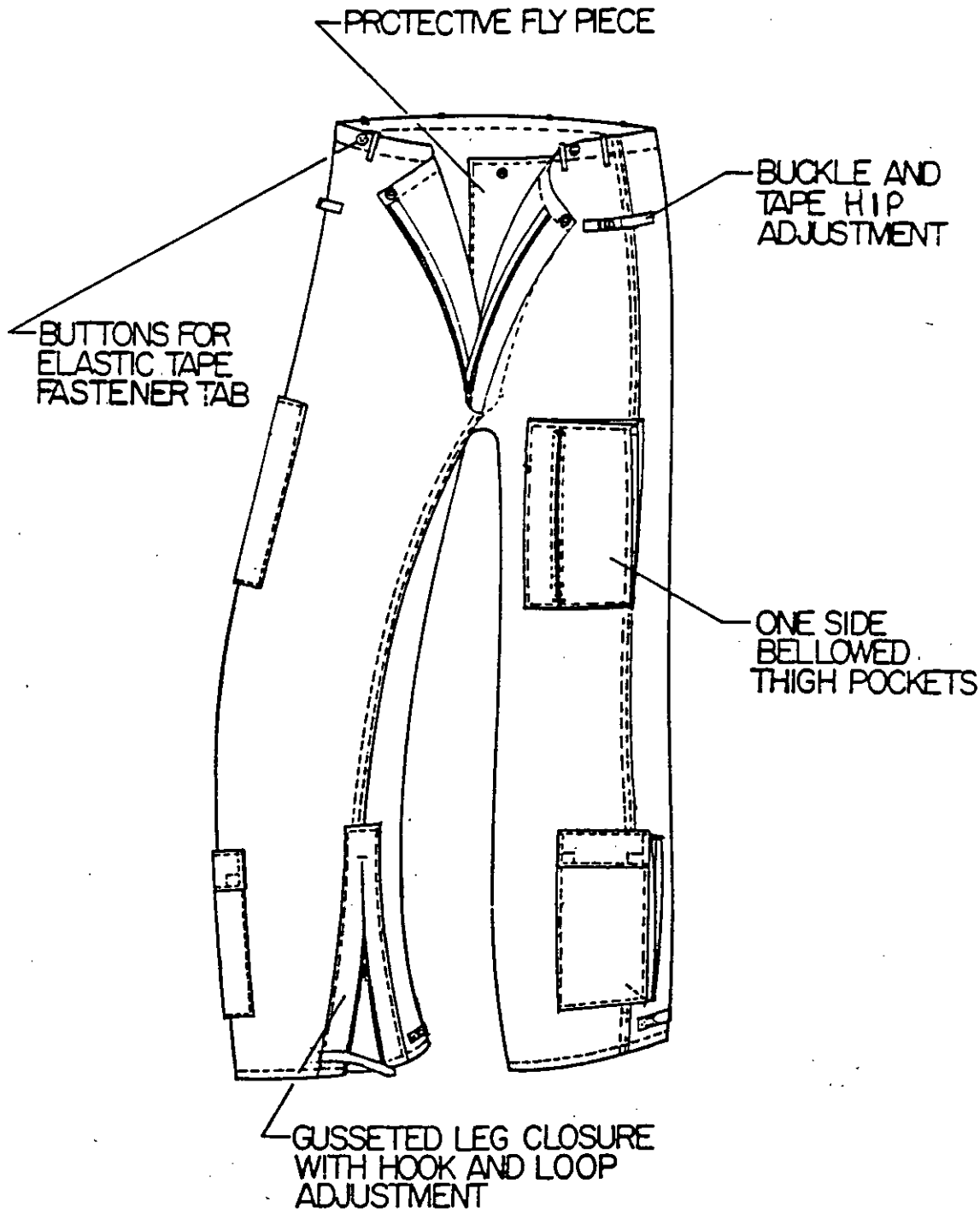


FIG. 3, TROUSERS, CHEMICAL PROTECTIVE: AIR CREW, FLAME RESISTANT 2-1-1855

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44356	1990 July 23

3. DOCUMENT TITLE
COAT AND TROUSERS, CHEMICAL PROTECTIVE, AIRCREW, FLAME RESISTANT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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